Less to Lose: How shorter and more flexible degrees shape university choice and earnings

Emma Scandolo

UCL

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Abstract

Higher-education systems differ in how long students must study before they earn their first university qualification. I study the Bologna Process, which moved the first qualification forward in two ways. In fields that already had a bachelor's and a master's, it shortened the bachelor's, so students could earn a degree sooner. In fields with a single long degree, it added a bachelor's partway through, as an intermediate qualification: the full path took as long as before, but students now earn a degree half way through. Using a regression discontinuity across birth cohorts and linked employer–employee data from Portugal, I find four main results. First, the reform raised university completion by 2.2 percentage points (8 percent) and hourly wages at ages 28 to 30 by 1.6 percent, with larger gains for women. Second, students moved towards both kinds of reformed field, and these shifts carried into the sectors where they work. Third, the intermediate qualification did not reduce the number of students completing the full degree: more students reached jobs that require it. Fourth, about two-fifths of the wage gain comes from working at higher-paying firms. Italian data show similar gains. A model of learning while enrolled, estimated only on the education responses, predicts a similar wage gain. In the model, shorter degrees attracted students by lowering cost. The intermediate qualification attracted them by lowering risk: it left students less to lose from trying a field.

1 Introduction

Higher-education systems differ substantially, both across and within countries. A key difference is how long students must study before they earn their first university qualification. A first degree in economics can take three years in Portugal, four in the United States and four or five in many Latin American countries. Even within the same country, fields can be structured very differently. The Portuguese economics student earns a bachelor's after three years and can then decide whether to continue studying. A medical student at the same university does not have the same flexibility: the first qualification comes only after six years. But what are the implications of choosing a system that lets students earn their first qualification earlier?

This question has received little attention, yet Europe offers a unique, large-scale natural experiment to study it. In 1999, 29 countries[1] launched the Bologna Process to make their higher-education systems more compatible and comparable (European Ministers of Education, 1999). At its core was a common sequence of degrees: a bachelor's of at least three years, followed by a master's and a doctorate. Most of western and central Europe chose a three-year bachelor's (Eurydice, 2024).[2] This moved the first qualification forward in two ways. In some fields, the reform shortened an existing first degree from four years to three. In others, it introduced a three-year qualification into what had been a single five-year degree, while completing the full programme still took five years. Advocates argued that shorter degrees and a more flexible structure would widen access. The European Commission, for example, pointed to large differences in tertiary attainment: only 21 percent of the EU working-age population held a tertiary qualification, compared with 38 percent in the United States (European Commission, 2005, p. 3). Critics worried that shorter degrees would mean less training and lower wages, because students would spend less time in university (Cappellari and Lucifora, 2009).

Two decades later, the debate remains open. Research on the reform has focused on enrolment and dropout, and its effects on careers remain little studied (Kroher et al., 2021). The question also matters beyond Europe. In Latin America, first degrees still take four to five years, and several countries, including Argentina and Chile, have recently shortened their degrees or are reviewing their length.[3] The answer is not

[1] Austria, Belgium, Bulgaria, Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, Netherlands, Norway, Poland, Portugal, Romania, Slovakia, Slovenia, Spain, Sweden, Switzerland, United Kingdom.

[2] Others, including Spain and Greece, opted for four years.

[3] Argentina set a four-year, 240-credit standard that new programmes must meet from 2027 (Ministerial Resolution 556/2025).

obvious ex ante.

Students incur costs and forgo earnings while enrolled. They also enrol with imperfect information about how likely they are to graduate and learn about their prospects only while studying. Because much of the return depends on obtaining a qualification (Hungerford and Solon, 1987; Jaeger and Page, 1996), costs accrue continuously while returns are lumpy. The timing of the first qualification shapes this investment in two ways, each involving a trade-off. First, shortening an existing degree lowers the cost of obtaining it and may attract new students who would otherwise not enrol. But completing the degree in less time may also mean receiving less training. Second, introducing an intermediate qualification into a long programme reduces the downside risk of uncertainty. Students who discover that the field is a poor match can obtain a qualification after three years rather than risk dropping out with no qualification. This may encourage new students to enrol, and some of them may discover that they are well matched and continue to the full degree. But it may also lead some students who would otherwise have completed the longer degree to stop after three years. Because these changes differed across fields, they may also affect what students choose to study. Their overall effects on education and earnings are therefore unclear.

These trade-offs motivate four questions. First, on net, how does moving the first qualification forward affect educational attainment and earnings? Do the gains from bringing new students into university outweigh the potential losses from shorter training and earlier stopping? Second, do students shift towards fields where the first degree became shorter, towards fields that gained an intermediate qualification, or both, and do these shifts carry into the sectors and occupations where they work? Third, does an intermediate qualification lead students to stop earlier, or can it encourage some to continue to the full programme? Can students' uncertainty about their match with a field explain this response? Finally, how much of any wage gain comes from the firms where people work?

To answer these questions, I exploit the rapid implementation of the Bologna Process in Portugal and Italy. I compare birth cohorts making their university choices immediately before and after the reform using a regression discontinuity design. Portugal is the main setting because of unusually rich linked employer–employee data. *Quadros de Pessoal* covers the Portuguese private sector and lets me observe workers'

In 2026, Chile's government and university rectors agreed to review the length of degree programmes.

education, wages, occupations, sectors, job titles and employers, and estimate firm wage premia. Because it covers only private-sector employees, and the reform might affect who works in the private sector, I check that the results hold when using the Portuguese Labour Force Survey and the Adult Education Survey, which are nationally representative. The Italian Labour Force Survey data then allow me to test the external validity of my findings. Because the Italian reform took effect several years earlier and Italian students enter university at a different age,[4] this comparison also helps address the concern that the Portuguese discontinuity captures contemporaneous macroeconomic conditions rather than the reform itself.

The analysis yields four main findings. First, on net, moving the first qualification forward raised educational attainment and wages, and both effects are larger for women. In Portugal, university completion rose by 2.2 percentage points, or 8 percent of the pre-reform mean. Average years of university rose by about 0.15 years, or 6 percent. Because post-reform cohorts start to work later, on average, they accumulate lower cumulative earnings in their 20s. By their early thirties, this shortfall has closed, and hourly wages at ages 28 to 30 are 1.6 percent higher for post-reform cohorts. Italian data show similar gains, concentrated among women. The gains from more students completing university therefore outweighed any losses from shorter training and earlier stopping. Second, students shifted towards both types of field restructured by the reform. Among women, completion of degrees that gained an intermediate credential, like engineering, rose by about a third, and completion of degrees that shortened the first degree to three years, like economics, rose by about a fifth. Degrees in unchanged fields, like healthcare fields, fell by more than a quarter. Men show the same pattern, with smaller changes. These shifts carry into the labour market. Post-reform cohorts are more likely to work for engineering and IT firms as well as in the finance and banking sector, and less likely to work in health. The extra graduates work in the sector that matches their field at rates somewhat below, but close to, those of earlier graduates, which suggests that the labour market largely absorbed them. Third, the intermediate qualification did not reduce the number of students who completed the full programme; it raised it. Engineering is the clearest case: employment rose in engineering jobs that require the full five-year programme, which implies that more students completed it. Finally, where people work explains part of the wage gain. An AKM decomposition (Abowd et al., 1999) shows that around 40 percent of the wage gain is

[4]Students typically enter university at 19 in Italy and at 18 in Portugal.

accounted for by workers entering firms with higher estimated wage premia.

I interpret these responses through the cost and the risk of the investment. Where the first degree was cut from four years to three, the reform lowered its cost. In engineering the full programme still took five years, but students who left after three years now left with a qualification instead of nothing. This did not reduce uncertainty; it reduced its downside, and so made engineering less risky to try. To test this interpretation, I build a simple model of degree choice. Students hold a belief about their match with each field, learn about it from their own progress, and can leave with whatever qualification they have reached. I assume that employers absorb the extra graduates at unchanged wages, as the evidence on where they work suggests. I estimate the model on the education responses alone. Without targeting wages, the model predicts a wage gain of 1.8 percent, close to the 1.6 percent in the data. In the model, the intermediate qualification draws in students who are unsure of their match. Some of them turn out to be well matched and complete the five years, and few students who would have completed five years stop at three. A counterfactual separates the two channels: every student learns her match at the end of her first year. The intermediate qualification is then worth almost nothing. The year saved by a shorter first degree keeps two-thirds of its value. Shortening the first degree attracted students by lowering the cost of a degree, and uncertainty made that saving worth more; the intermediate qualification attracted them only by lowering the risk of trying one.

Existing research on the Bologna Process has focused mainly on school leavers' enrolment decisions and on dropout or on-time graduation conditional on enrolment (Cappellari and Lucifora, 2009; Cardoso et al., 2008; Horstschräer and Sprietsma, 2015; Bondonio and Berton, 2018). Evidence on earnings remains limited, mainly descriptive (Kroher et al., 2021, for a review), and generally conditions on university completion (Bosio and Leonardi, 2010; Glauser et al., 2019; Suleman and Figueiredo, 2020). A second strand estimates how the length of a degree affects graduates' earnings, with mixed results: cutting coursework lowered wages by 13 to 16 percent at Colombia's top university (Arteaga, 2018), a one-year cut in Russia had no average effect (Avdeev, 2025), and an extra honours year in Singapore raised earnings by about 12 percent (Gong and Pan, 2023). However, these studies compare graduates and hold enrolment fixed. This paper makes four contributions. First, it provides, to my knowledge, the first unified causal account of how moving the first qualification forward changes both education and labour-market outcomes. By estimating effects across

whole cohorts rather than only among university entrants or graduates, I capture the combined responses in enrolment, graduation, years of study, field choice and earnings. The results show that the reform did more than change the qualifications held by existing students: it changed who acquired a university education, what they studied and where they subsequently worked. Second, the paper changes how the reform should be understood. Existing work treats a shorter degree as cheaper but providing less training. I show that an intermediate qualification also lowers the risk of trying university. This connects the paper to models of schooling as experimentation, in which students learn about their match while enrolled (Manski, 1989; Stinebrickner and Stinebrickner, 2012; Stange, 2012; Arcidiacono et al., 2025), and to stepping-stone models in which an earlier qualification creates option value (Trachter, 2015). Existing work typically quantifies the value of options already present in the education system. I instead use a policy-induced change that created a new option in the data. Third, the paper broadens the empirical literature on stepping stones in higher education. Existing evidence comes mainly from US community colleges, where the intermediate qualification is provided by a separate institution (Rouse, 1995; Jepsen et al., 2014; Stevens et al., 2019; Mountjoy, 2022). Bologna instead inserted an intermediate qualification within the university programme itself. This setting shows that a stepping stone need not divert students from longer qualifications: it can draw in new students and increase progression to the full degree. Finally, I address an important limitation of the US community-college literature, which relies largely on unemployment-insurance records. These records measure quarterly earnings, which mix pay with hours worked, and provide little information about the jobs and employers through which earnings gains arise. The Portuguese linked employer–employee data record hourly wages and hours separately, so I can tell higher pay from longer hours. They also identify each worker's occupation, sector and employer, so I can trace the wage effect through these channels and through firm wage premia. In this respect, the paper builds on Portugal et al. (2024), who use the same data to decompose the returns to compulsory schooling in Portugal. I extend this approach to higher education and show what part of the wage effect is attributable to firms.

The rest of the paper is structured as follows. Section 2 describes the institutional setting. Section 3 presents the data sources and explains how the samples are built. Section 4 sets out the empirical strategy. Section 5 presents the results, and Section 6 presents the model. Section 7 concludes.

2 Institutional Setting

The Bologna Declaration of 1999 committed European countries to a common structure of higher education: a first cycle of at least three years, followed by a master's. Countries adopted the reform at different times and in different forms (Cardoso et al., 2008). Portugal and Italy both chose a three-year first degree followed by a two-year master's, and both countries gave a short time span of around two years to universities to fully implement the reform.

Timeline. Both roll-outs were fast (Figure 1). In Portugal, Decree-Law 74/2006 let programmes convert from 2006/07 and required them to by 2008/09; 43 percent converted in the first year and about 90 percent by 2007/08 (Cardoso et al., 2008), so the 2007 applicants are the first cohort to face an almost fully reformed system. In Italy, the Ministry of Education released guidelines on how to implement the Ministerial Decree 509/1999 in August 2000. Universities whose pre-scheduled faculty meetings took place in September 2000 could implement the reform from the academic year 2000-01. In practice, most universities adopted the new regime in 2001/02 (Cappellari and Lucifora, 2009; Bondonio and Berton, 2018).

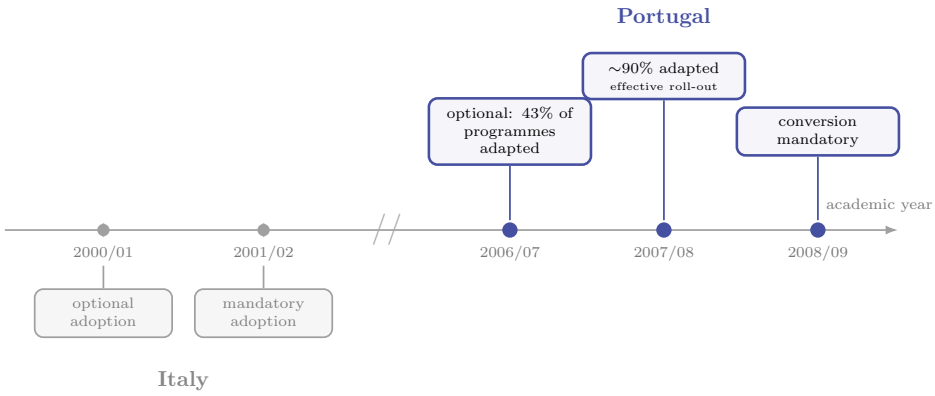


Figure 1: Implementation of the reform in Portugal and Italy

Notes: Sources: Decree-Law 74/2006; Ministerial Decree 509/1999.

Restructuring by field. The reform shortened the time to a first credential in two different ways (Figure 2). First, where a field was organised as a single long degree, the reform introduced a new intermediate degree after three years where none had existed before. This was the case in every reformed field in Italy, where the four- to six-year *laurea* had no degree inside it and no master's above it; it became a three-year

laurea followed by a two-year *laurea specialistica*. In Portugal it was the case in university engineering and computing, which lasted five years; this was replaced either by a three-year bachelor's and a two-year master's, or an integrated master's that by law awards the first degree after three years (Decree-Law 74/2006, art. 19). The path to the profession still takes five years, but a student who leaves after three now leaves with a degree instead of nothing. Second, where a field was already split into a first degree and a master's, the reform shortened the first degree. This happened only in Portugal. The *licenciatura* in economics, management, the sciences, the social sciences and the humanities lasted four years (five at some institutions) and could be followed by a two-year master's, which about one graduate in ten took; the reform cut it to three years. Economics at the University of Porto illustrates what was removed from an economics *licenciatura*. Between the four-year plan of 2004/05 and the three-year plan of 2008, the degree lost Microeconomics III, Mathematics III, Econometrics II, and four of its twelve applied field courses (international monetary economics, innovation, regional economics and economic policy), while several of the remaining field courses shrank from six to four credits and the number of management and finance courses rose from six to nine.[5] The first cycle became the old degree minus most of its final year, which moved into the master's. Some fields maintained the same structure after the reform. Medicine and law kept long degrees of five to six years in both countries. Nursing stayed four years in Portugal; in Italy it was already a three-year university diploma, but became a recognised university degree followed by a master's.

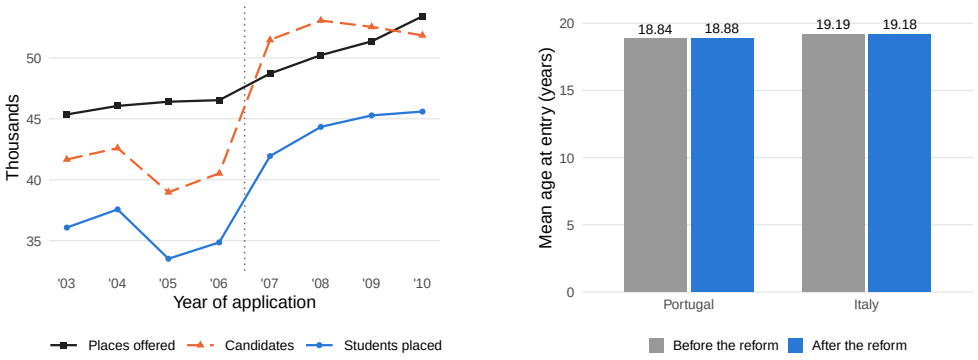


Figure 2: Years to the first university credential before and after the reform, by field

Notes: Statutory years to the first degree. (+2): pre-existing master's. Blue: new intermediate credential; orange: shorter first-cycle degree. Polytechnic and short-diploma programmes not shown.

[5]Deliberations 396/2004 and 397/2004 (*Diário da República*, 26 March 2004) and the faculty's official Bologna study plan (2008). The removed material reappears as options or in the new master's in economics. Cardoso et al. (2008) describe the new curricula more generally as compressed versions of the old ones.

Enrolment: demand, supply and age at entry The increase in enrolment was driven by demand, not by supply of places. In Portugal each institution proposes the number of places per programme and candidates are allocated through a national contest. Places grew by 5 percent in 2007, while candidates rose by 27 percent (Figure 3a); the extra students filled places that already existed, since in 2006 one place in four had gone unfilled. In Italy most programmes had no cap on entry, so enrolment was not rationed by places at all; first-time entrants rose by 12 percent in 2001/02 and stayed about 18 percent above their 1998–2000 level. The reforms did not change the age at which students enter (Figure 3b).



(a) Places, candidates and students placed, Portugal

(b) Mean age at first entry

Figure 3: Enrolment: places versus demand, and age at entry

Notes: (a) Data from DGES. (b) Data from Eurostat.

3 Data

3.1 Data sources

I combine one administrative dataset and three surveys. *Quadros de Pessoal* is the backbone of the Portuguese analysis: it follows every private-sector employee over time and records education, field of study, occupation, job title, pay and employer for the full cohorts on both sides of the reform. Two Portuguese surveys fill the gaps it leaves. The Labour Force Survey covers the people *Quadros de Pessoal* never sees—public-sector workers, the self-employed and those out of work—and the Education and Training Survey records who entered higher education and who left without a degree. The Italian Labour Force Survey supplies data for the Italian analysis, which I use to test whether the Portuguese patterns travel elsewhere.

Quadros de Pessoal. *Quadros de Pessoal* (QP) is a linked employer–employee data set collected every year by the Portuguese Ministry of Labour from all private-sector establishments with at least one employee. Reporting is compulsory, and each establishment reports its workforce as of October. QP therefore covers the full population of workers and firms in manufacturing and private services; it excludes civil servants, the self-employed and domestic workers. Public hospitals are nonetheless largely covered as most of them are public enterprises (*Entidades Públicas Empresariais*, EPE), which report to QP. For each worker I observe sex, birth date, the highest completed level of education and its field, a four-digit occupation code, the job title within the applicable collective agreement (*categoria profissional*), the worker's hierarchical level within the firm (*nível de qualificação*), normal monthly hours, and monthly base pay, regular supplements and overtime pay. Workers and firms carry anonymised identifiers, so I can follow a worker across employers and years. I use the waves from 2000 to 2023; no worker file exists for 2001.

The QP education variable measures well whether a worker holds a university degree, but it cannot reliably tell one level of degree from another. Before 2006 the education classification has no separate code for the master's degree, which is pooled with the *licenciatura*; meaning I can distinguish master's from first-cycle degrees only from 2006 onwards. Splits of the degree effect by level and length in QP are therefore less reliable than the effect on holding any degree. For years of university, I rely on a proxy calculated as the age an individual first appears in QP minus 18 for those who attended university, and 0 otherwise. I validate this imperfect measure using the Portuguese LFS, which record the age at which a person left education, and on the Italian LFS, which additionally codes the length of the degree directly.

Portuguese Labour Force Survey (LFS). The *Inquérito ao Emprego* is the quarterly household survey of Statistics Portugal (INE) and is representative of the resident population. Unlike QP it covers every employment status, so it measures educational attainment without conditioning on private-sector employment, or on employment at all. It records the highest completed level of education and its field, the age at which the respondent left education, employment status and monthly pay, with education codes close to those in QP. I use the quarterly files from 1998 to 2024.

Education and Training Survey (IEFA). The *Inquérito à Educação e Formação de Adultos* is INE's module of the European Adult Education Survey. I use its four waves, 2007, 2011, 2016 and 2022. It is the only Portuguese source that records whether a respondent ever enrolled in higher education and left without a degree; it also records the level and field of the highest degree and, from 2016, whether that degree was a post-Bologna degree. The samples are small—about 3,600 women and 3,500 men in the cohort window—so I use IEFA for enrolment and dropout rather than for precise effects on attainment.

Italian Labour Force Survey. The *Rilevazione sulle Forze di Lavoro* is the quarterly household survey of the Italian statistics office (ISTAT). I use the public-use quarterly files from 2009, the first year that reports year of birth, to 2020. The survey records the highest completed level of education and separates the two-cycle degrees created by the reform—the three-year *laurea* and the two-year *laurea specialistica*—from old-regime and single-cycle degrees. It also records the degree subject, the age at which the respondent left education, employment status, occupation and net monthly pay in the main job for employees, top-coded at 3,000 euros. I add ministry counts of students enrolled in two-cycle programmes by university and year, which identify the regions whose universities adopted the reform in 2000/01.

3.2 Sample construction

The design compares people who reached the age of university entry just before and just after the reform, so the unit of comparison is the entry cohort. I build every sample in two steps. I first assign each person to the year in which they faced the decision to enrol. Then, I measure that year relative to the year the reform reached them.

Portugal. Portuguese students apply to university at the end of secondary school, usually at 18, so I assign each person to the calendar year in which they turned 18, $c_i = \text{year of birth} + 18$. The reform reached almost every programme within one year: 43 percent of programmes converted in 2006/07 and about 90 percent by 2007/08 (Section 2). The cohort that turned 18 in 2007, born in 1989, is therefore the first to apply to an effectively fully reformed system, and the running variable is $Z_i = c_i - 2007$. The 2006 cohort applied when fewer than half of the programmes had converted. It is only partly exposed, so I exclude it rather than

assign it to either side. I keep the six cohorts on each side of this gap, those turning 18 in 2000–2005 and in 2007–2012 (born 1982–1987 and 1989–1994). The same twelve cohorts define every Portuguese sample.

Italy. In Italy the reform did not reach every student in the same year, so the cutoff depends on where a person studied. To date it, I use ministry counts of students enrolled in two-cycle and in old-regime programmes at each university from 2000/01, aggregated to the region in which the university is located. In 2000/01 only four regions had a non-negligible share of students in two-cycle programmes: Piemonte, Valle d'Aosta, Friuli-Venezia Giulia and Sardinia. I call these the early adopters, which converted in 2000/01, and the rest of the country the late adopters, which converted in 2001/02. The Labour Force Survey does not record where a respondent studied, so I assign each respondent to the adoption year of their region of residence. This is a close proxy, because regional mobility in Italy is low: 84 percent of graduates live in the region where they studied (Bondonio and Berton, 2018). Italian students usually enter university at 19, so with $c_i = \text{year of birth} + 19$ the running variable is

$$Z_i = \begin{cases} c_i - 2000 & \text{if } i \text{ lives in an early-adopting region,} \\ c_i - 2001 & \text{if } i \text{ lives in a late-adopting region.} \end{cases} \tag{1}$$

The survey data confirm the classification: the share of respondents with a two-cycle degree jumps one cohort earlier in the early-adopting regions than in the rest of the country (Appendix Figure 11).

The resulting samples contain 12.6 million QP worker-years from 1.8 million workers, 310,355 respondents in the Portuguese Labour Force Survey, about 7,100 IEFA respondents and 269,026 respondents in the Italian survey. Appendix B.1 defines the education and labour-market outcomes.

4 Empirical Strategy

I compare birth cohorts that reached the age of university entry just before and just after the reform. Because the reform reached almost every university programme within a single year (Section 2), whether a student faced the old or the new degree structure depended on the year they were born, not on anything they chose. This section sets out the regression discontinuity design. Appendix C reviews the evidence that supports the design, and describes the decomposition I use to split the wage effect into a firm and a worker component.

Let c_i be the calendar year in which person i turns 18, the usual age of application in Portugal. The running variable is $Z_i = c_i - 2007$, and the treatment indicator is $D_i = \mathbf{1}[Z_i \geq 0]$: the cohort turning 18 in 2007 is the first to apply to an almost fully reformed system. The 2006 cohort applied when 43 percent of programmes had converted and is only partly exposed, so I drop it. I keep six cohorts on each side, those turning 18 in 2000–2005 and in 2007–2012.

For an outcome Y_{it} of person i observed in year t at age a_{it}, I estimate

$$Y_{it} = \alpha_0 + \alpha_1 D_i + \alpha_2 Z_i + \alpha_3 D_i Z_i + g(a_{it}) + \varepsilon_{it}, \tag{2}$$

where $g(\cdot)$ is a quadratic in age. The coefficient of interest is α_1, the jump in the outcome at the cutoff. The terms in Z_i allow a separate linear cohort trend on each side, so α_1 compares the first reformed cohorts with what the trend of the pre-reform cohorts predicts for them. Outcomes are the education measures and labour-market outcomes defined in Appendix B.1.

I estimate equation (2) on cohort means: I remove the age profile, average each outcome within a cohort and fit a line on each side of the cutoff to the twelve cohort points. Standard errors therefore treat the cohort, not the worker, as the unit of observation (Appendix C.2). For wages, I also remove calendar-year effects, as explained in Appendix C.4.

The design requires that, without the reform, outcomes would have continued smoothly across the 2007 cutoff. Four pieces of evidence support it. First, people cannot choose the year they were born, and the number of workers per cohort in *Quadros de Pessoal* shows no break at the cutoff (Figure 12, Appendix C.3). Second, students who turned 18 before 2007 did not wait to enter the new degrees. If they had, more of them would have worked between leaving secondary school and graduating from university; the discontinuity in that share is 0.1 percentage points on a base of 10.5 percent, and it is close to zero for both women and men (Table 9, Appendix C.3). The mean age at first entry also did not change (Figure 3b), which supports assigning exposure by the year a cohort reaches the usual age of entry. Third, *Quadros de Pessoal* sees only private-sector employees, so a reform that changed who works in the private sector would show up as a change in the sample. I validate that every education result replicates in the Labour Force Survey, which covers the whole population. Fourth, the jump is specific to 2007. When I move the cutoff to cohorts where nothing changed, using only pre-reform or only post-reform cohorts, the placebo jumps are smaller than

the actual one for completion, years of university, the three field groups and wages. The one exception is men's years of university, where one placebo is as large as the small actual jump but of the opposite sign (Appendix D).

For Italy I estimate the same equation. However, in Italy I do not use a donut setting because the running variable is the year a person turns 19 minus the year universities in their region of residence adopted the reform, 2000 or 2001 (Section 3.2).

5 Results

This section follows the four questions of the introduction. Section 5.1 asks whether, on net, the reform raised educational attainment and earnings. Section 5.2 asks which fields students moved towards, and whether these moves carried into the sectors and occupations where they work. Section 5.3 asks whether the new three-year qualification led students to stop early or whether they continued to the full degree. Section 5.4 decomposes the wage gain. The first three subsections end with the Italian evidence. Throughout, I report effects separately for women and men, since the reform affected them differently at almost every margin. Each estimate is the coefficient α_1 in equation (2), estimated as described in Section 4, and "baseline" is the pre-reform mean of the outcome; the note to each table and figure gives its exact definition.

5.1 More graduates, more years of university and higher pay

University completion and years of university. The reform raised university completion. Figure 4 plots the share of each cohort whose highest completed level is a university degree, together with average years of university in the population, counting zero for those who never graduated. Both series jump at the first reformed cohort, and the jump persists in the cohorts that follow. Among women the share with a degree rises by 2.9 percentage points, from a baseline of 34 percent, or 8.6 percent; among men it rises by 1.4 percentage points from 20 percent, or 7 percent. The effect on women is twice as large in levels; the difference is significant at the 10 percent level ($p = 0.05$). Because the reform shortened the first degree, more graduates need not mean more schooling. Students who would have completed a four- or five-year degree could instead stop after three. On net, however, schooling rises. Average years of university rise by

0.23 years for women and 0.07 years for men, about 8 and 4 percent of the baseline. Other data sources give the same result (Appendix Table 13). This matters because *Quadros de Pessoal* observes only private-sector employees, so a reform that changed who works in the private sector could produce a jump within QP without any change in the population. The LFS covers the whole resident population. It gives a larger effect on women's completion, 3.7 percentage points on a baseline of 23 percent, and a similar effect on men's, 1.2 percentage points. Its effects on years of university are close to those in QP. IEFA points in the same direction, but with about 3,500 respondents of each sex its estimates are too imprecise to distinguish from zero.

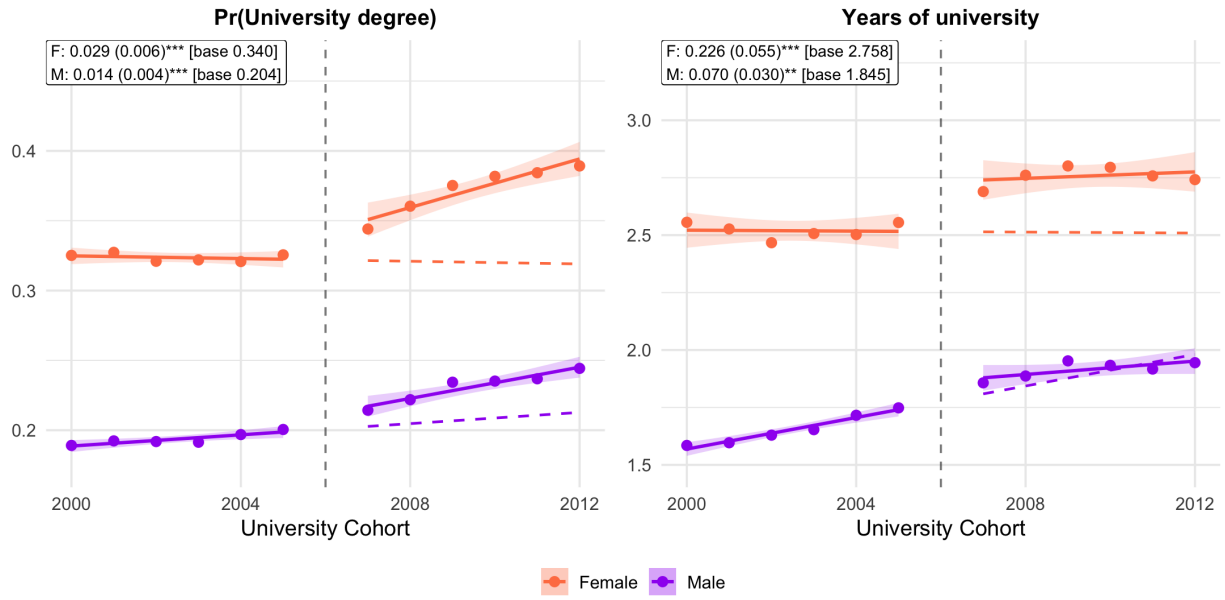


Figure 4: University completion and years of university, by cohort

Notes: Quadros de Pessoal. Dots are cohort means; solid lines are linear fits on each side of the cutoff with 95 percent confidence bands; dashed lines extend the pre-reform fit. Cohorts are defined by the year a person turned 18; the 2006 cohort is excluded. Left panel: share whose highest completed level is a university degree. Right panel: years of university, zero for non-graduates. Boxes report the RD estimate, its standard error and the baseline, the mean of the outcome in the pre-reform cohorts 2000–2005.

Hourly wages at ages 28 to 30. Post-reform cohorts earn higher hourly wages. I measure the wage at ages 28 to 30, when most people who went to university have finished their studies and started working, and remove calendar-year effects (Appendix C.4). At the cutoff, the log hourly wage rises by 1.6 percent (Table 1, column 1). The effect is 2.2 percent for women and 1.0 percent for men, and the difference between them is statistically significant. The larger effect for women matches their larger gains in completion and

years of university. Wider age windows give the same result (Appendix C.5).[6] I find no significant effect on employment at these ages, for women or for men, measured from LFS data (Table 1, column 2).

Table 1: Effect of the reform on log hourly wages and employment at ages 28–30

	Log hourly wage QP (1)	Employed LFS (2)
All	0.016***	0.009
	(0.003)	(0.021)
Baseline	1.513	0.759
Women	0.022***	0.027
	(0.003)	(0.026)
Baseline	1.478	0.748
Men	0.010***	−0.011
	(0.003)	(0.018)
Baseline	1.546	0.771
p-value, women = men	0.01	0.22
Cohorts, pre / post	6 / 6	6 / 6
Calendar years	2010–2023	2010–2023

Notes: RD estimates at the 2007 cohort. Cohort effects net of age and calendar-year effects, estimated on cohort × age × year cells weighted by observations (the whole sample adds a sex dummy); then a line on each side of the cutoff on the cohort points, the coefficient being the jump at the 2007 cohort. Cohorts 2000–2012, 2006 excluded; the 2012 cohort is observed at ages 28–29 only. Column 1: *Quadros de Pessoal*, workers aged 28–30, main job, workers in the largest connected set; log real hourly wage (monthly hours and earnings winsorised at the 99th percentile within year); baseline: mean log hourly wage of the 2005 cohort at ages 28–30, in euros of 2015–2017. Column 2: LFS, everyone aged 28–30, unweighted; employed (ILO definition); baseline: share employed in the 2005 cohort at ages 28–30. Observations (women / men): QP 1.09 / 1.20 million worker-years; LFS 20,784 / 21,494 person-quarters. Other ages and another way of removing calendar time: Appendix Table 11. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Cumulative earnings. The wage at 28 to 30 does not show what post-reform cohorts gave up while they studied longer. Measuring wages at younger ages would not show it either. At 23 or 25, some post-reform students are still in university, so a wage effect at those ages would mix the effect on pay with the later start of work, and the two could not be told apart. Cumulative earnings address this. I sum each person's annual earnings from age 16 up to a given age, counting zero in years they are not observed in QP, and estimate the RD at each age. The sample is everyone who appears in QP at some point, so these totals are still conditional on eventually working in the private sector. They do not measure lifetime earnings in the whole cohort. They show two things: how much income post-reform cohorts forgo while they study, and whether the higher wages they earn once they start working make up for it. Figure 5 plots the estimates. From then on post-reform cohorts earn more each year, and the gap closes around age 30 for men and 31 for women. The higher wages thus offset the earnings forgone while studying by the early thirties.

[6] Mountjoy (2022) averages earnings over the same ages.

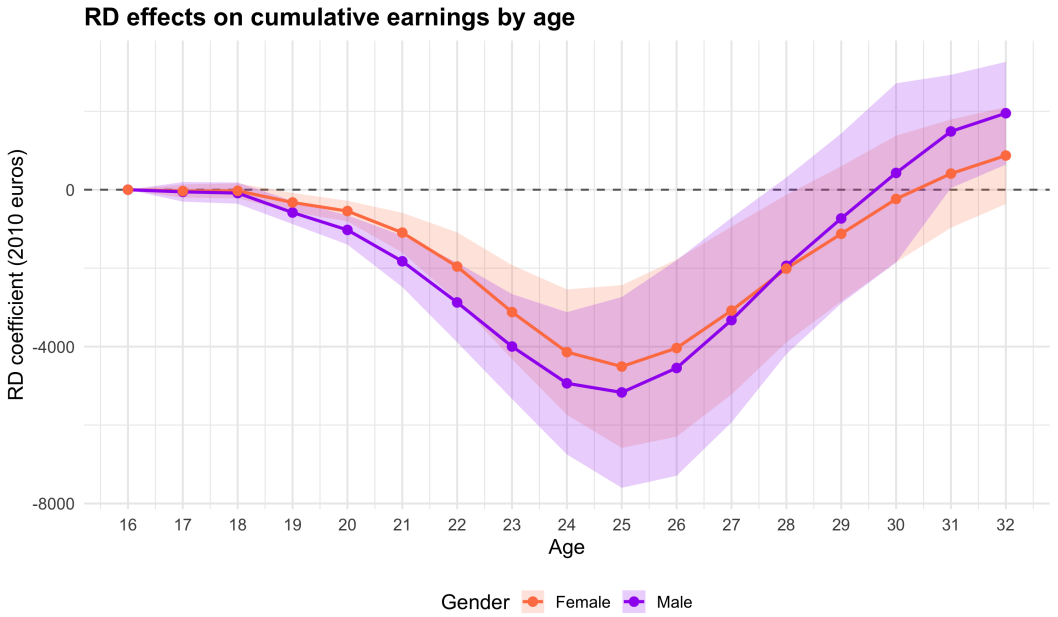


Figure 5: Effect of the reform on cumulative earnings, by age

Notes: Quadros de Pessoal. Each point is a separate RD estimate of cumulative annual earnings from age 16 up to the age shown, in 2010 euros (consumer price index), with years not observed in QP counted as zero, and a 95 percent confidence band. Sample: everyone observed in QP at least once. Annual earnings are October monthly earnings times twelve, summed over jobs. RD on cohort-by-sex means, linear on each side, cohorts 2000–2012 (2006 excluded). A cohort enters at an age only if it is observed at that age by 2023, so at 31 the post-reform side has four cohorts and at 32 three (2007–2009). Appendix B.1 describes the construction.

Italy. Italy reformed its universities along the same lines six years earlier, and its students enter at a different age, so a Portuguese jump caused by something else that happened to the 2007 cohort would not reappear there. I therefore repeat each of the first three sets of results on Italian data. Table 2 shows the main ones. The share of Italian women holding a university degree rises by 1.8 percentage points at the cutoff, and the share of men by 0.8 percentage points, compared with 2.9 and 1.4 percentage points in Portugal (column 1). Average years of university rise by 0.17 years for women, or 9 percent, and by 0.07 years for men (column 2). Italian wages show the same gender pattern as Portuguese ones. Among Italian women aged 30 to 36, the log net monthly wage rises by 2.1 percent (column 3). Among men it does not change, and the difference between women and men is statistically significant. The probability of being employed does not change for either (column 4). The Italian wage effect for women is the same size as the Portuguese one, although it is measured with a different wage concept, in a different country and six years earlier.

Table 2: Effect of the reform on university attainment, wages and employment in Italy

	Pr(university) (1)	Years of university (2)	Log net monthly wage (3)	Employed (4)
All	0.014***	0.125***	0.008*	−0.002
	(0.003)	(0.026)	(0.005)	(0.004)
Baseline	0.227	1.626	7.029	0.698
Women	0.018***	0.169***	0.021***	0.002
	(0.005)	(0.039)	(0.007)	(0.005)
Baseline	0.273	1.900	6.910	0.608
Men	0.008*	0.069*	0.001	−0.003
	(0.004)	(0.035)	(0.005)	(0.005)
Baseline	0.179	1.340	7.134	0.791
p-value, women = men	0.17	0.06	0.02	0.55
Observations	269,026	269,026	142,599	269,026

Notes: RD estimates at the year of adoption. Italian Labour Force Survey, Italian citizens aged 30–36. The running variable is the year a person turned 19 minus the adoption year in their region; cohorts −4 to 3. Individual-level regression with a line on each side of the cutoff, the coefficient being the jump at the cutoff. Column 1: holds a university degree. Column 2: years of university, zero without a degree and age at the end of study minus 19 with one. Column 3: log net monthly pay in the main job, top-coded at 3,000 euros, employees only. Column 4: employed. Baseline: mean of the pre-reform cohorts. The p-value comes from the pooled regression with every term interacted with a female dummy. Observations (women / men): columns 1, 2 and 4, 137,193 / 131,833; column 3, 66,675 / 75,924. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.2 Fields of study, sectors and occupations

Degrees by field group. The reform did not treat all fields alike, and students responded to the differences. I group fields by what the reform did to the time to a first credential (Figure 2): fields where a five-year degree gained a three-year exit ("5 to 3+2", mostly engineering and the sciences); fields where a four-year first degree became three years ("4+2 to 3+2", economics and management, the social sciences, the humanities and the arts); and fields with no usable early exit ("no change", health, including nursing, and law). The outcome is the share of the whole cohort holding a degree in each group, so it combines students drawn into university with students who would have enrolled anyway but chose a different field. Figure 6 shows three results. First, degrees in fields that gained a three-year exit rise by 2.1 percentage points for women, about a third of their baseline of 6 percent, and by 1.3 percentage points for men, about 12 percent of their baseline of 10 percent. Second, degrees in fields where the first degree became shorter rise by 2.6 percentage points for women, a fifth of their baseline, and by 0.6 percentage points for men. Third, degrees in the unchanged fields fall, by 2.6 percentage points for women and 0.6 percentage points for men, in both cases more than a quarter of the baseline. Appendix Figure 14 splits the groups into fields. Most of the rise is in engineering and computing and in economics and business. Engineering and computing degrees rise by 1.7 percentage points for women, from a baseline of 3.9 percent, and by 1.1 percentage points for men,

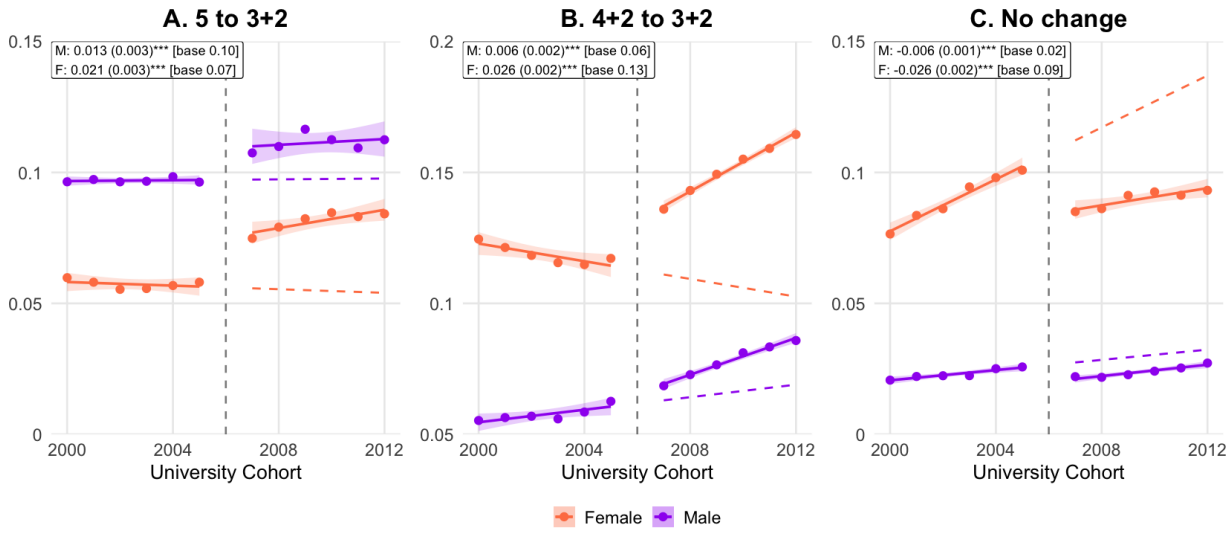


Figure 6: University degrees by what the reform did to the field, by cohort

Notes: Quadros de Pessoal. Share of the cohort whose highest completed level is a university degree in each group of fields. Panel A: fields where a five-year degree became three plus two (STEM). Panel B: fields where a four-year first degree became three years. Panel C: fields with no usable early exit (health and law). See the notes to Figure 4.

from 8.8 percent. This is four-fifths of the rise in STEM for women and nine-tenths for men; degrees in the sciences and mathematics rise by only 0.4 and 0.1 percentage points. Economics and business degrees rise by 1.1 percentage points for women and 0.4 percentage points for men. The social sciences, the humanities and the arts, and education each rise by 0.5 to 0.9 percentage points for women and by less for men. The fall is in nursing and the other health degrees, which drop by 2.7 percentage points for women and 0.6 percentage points for men. Medicine and law barely move. Nursing stayed a four-year degree in Portugal and so became long relative to the alternatives. The rise in the two reformed groups is larger than the overall rise in graduates, so the reform did more than add students: it also moved students who would have graduated anyway out of the unchanged fields. Other data sources show the same pattern (Appendix Table 14). In the LFS, degrees in the fields whose first degree became shorter rise by 3.7 percentage points for women and 0.9 percentage points for men, and health degrees fall, by 1.0 and 0.3 percentage points. Degrees in the fields that gained a three-year exit rise by 0.8 percentage points for women; for men the effect is small and not different from zero. The IEFA estimates have the same signs for women but are too imprecise to distinguish from zero. Ministry records of applications and first-year students show the same shift at entry, in public and private institutions, before anyone reaches the labour market (Appendix G).

Sectors and occupations. The shift across fields carries into the jobs people hold. Figure 7 compares, for one field of each type, the jump in the share of the cohort holding a degree in the field with the jump in the share working in the matching sector and in the matching occupation. This figure measures each outcome among worker-years at ages 28 to 32, and its engineering degrees exclude computing, so the degree effects differ slightly from those in Figure 6. For women, all three move together. Degrees in engineering rise by 1.6 percentage points, work in IT and engineering firms by 0.7 points and work in engineering occupations by 1.2 points. Degrees in economics and business rise by 0.9 points, and work in finance and business services and in business occupations by 0.7 to 0.8 points. In health, the share with a degree falls by 3.3 points, work

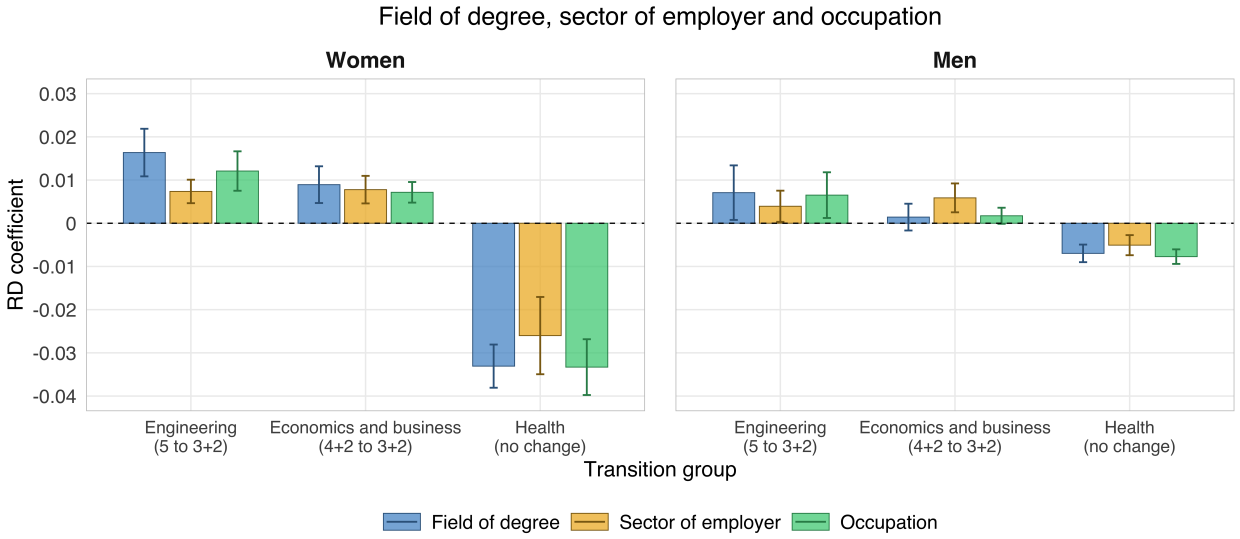


Figure 7: Field of degree, sector of employer and occupation, by what the reform did to the field

Notes: *Quadros de Pessoal*, worker-years at ages 28–32 in 2008–2023, cohorts 2000–2012 (2006 excluded). RD estimates at the 2007 cohort with age and calendar-year effects, linear on each side, with 95 percent confidence intervals. Each outcome is a share of all the cohort's worker-years; none conditions on holding a degree or on the sector. Engineering degree: engineering, manufacturing and processing, and architecture and construction; sector: CAE 62, 63 and 71; occupation: ISCO-08 214, 215, 251, 252 and 311. Economics and business: sector CAE 64–66, 69 and 70; occupation ISCO-08 241–243. Health: sector CAE 86; occupation ISCO-08 221–226 and 321–325.

in health firms by 2.6 points and work in health occupations by 3.3 points. Men show the same pattern with smaller changes, except that work in finance and business services rises by 0.6 points although their economics and business degrees barely change. Most of the extra graduates found work in their own fields, though at somewhat lower rates than earlier graduates. Among the graduates the reform added, the share working in the sector that matches their field is 24 percent in business, against 33 percent of pre-reform graduates, 24 against 28 percent in teaching, and 15 against 21 percent in engineering, architecture and

computing. In health, where the reform reduced the number of graduates, 74 percent of the graduates lost would have worked in health, against 67 percent of pre-reform graduates. These shares are ratios of two RD estimates and are imprecise. Appendix H measures the match directly at graduates' first jobs. In engineering, the share of new graduates whose first job matches their field does not change significantly. In economics and business, the occupation match does not fall, and the sector match is a few points lower but mostly not significantly so.

Italy. Figure 8 puts the two countries side by side, with Italian fields grouped by the same rule.

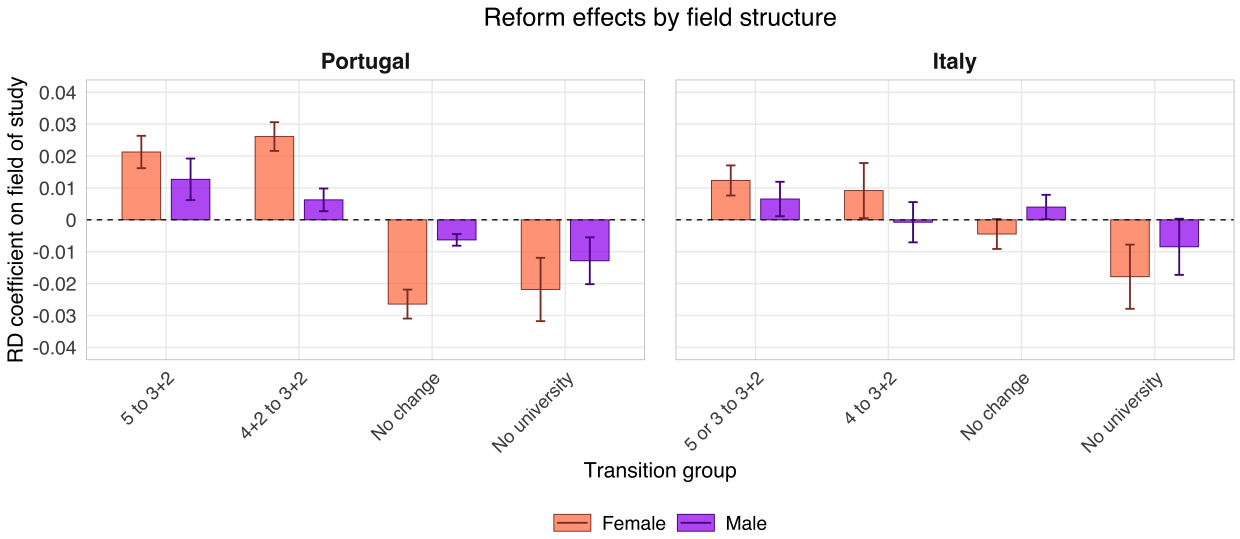


Figure 8: Effects on degrees by what the reform did to the field, Portugal and Italy

Notes: RD estimates with 95 percent confidence intervals. Portugal: *Quadros de Pessoal*, cohorts defined by the year a person turned 18. Italy: Labour Force Survey, citizens aged 30–36, cohorts defined by the year a person turned 19 relative to the regional adoption year. In Italy, the first group also includes the three-year *diploma universitario*, which became a first degree.

In Italy, degrees in the fields that gained a three-year exit rise by 1.2 percentage points for women and 0.7 percentage points for men; degrees in fields whose four-year degree became three plus two rise by 0.9 percentage points for women and do not change for men. The Italian field effects are about half the Portuguese ones, and in both countries they are larger for women. The second group, however, is not a direct counterpart of its Portuguese equivalent. In Italy the four-year *laurea* had no master's above it, so the reform did two things at once: it brought the first degree forward by a year, and it made that degree the first step towards a five-year qualification. The old four-year degree was also equated in law with the new master's, not with the three-year degree (Appendix A.2). A student who wanted a qualification of the old

standing therefore faced a longer path after the reform. This may explain why the response in this group is smaller in Italy than in Portugal, and absent among men.

5.3 The intermediate qualification and the full degree

Portugal. Critics worried that students would spend less time in university, and so receive less training and earn lower wages. In engineering, this would happen if students took the new exit after three years instead of completing five. Engineering provides a direct test. In Portugal the title of engineer, and the acts reserved for it, require the full five-year cycle, so an engineering occupation reserved for full-cycle graduates can only be reached by students who continued past the third year (Appendix A.2). If the reform had mainly diverted engineering students to the three-year exit, employment in these occupations would not rise. It rises. Figure 9 shows that engineering degrees rise by 1.3 percentage points for women and 0.9 percentage points for men. These are smaller than the effects in Appendix Figure 14 because here engineering excludes computing and uses only the field that workers report. Employment in engineering occupations rises by 0.9 and 0.5 percentage points, and employment in the engineering occupations that require the five-year degree rises by 0.6 percentage points for both sexes, roughly half of the women's baseline and a fifth of the men's. Pooling women and men, the reform raised engineering degrees by 1.1 percentage points on a base

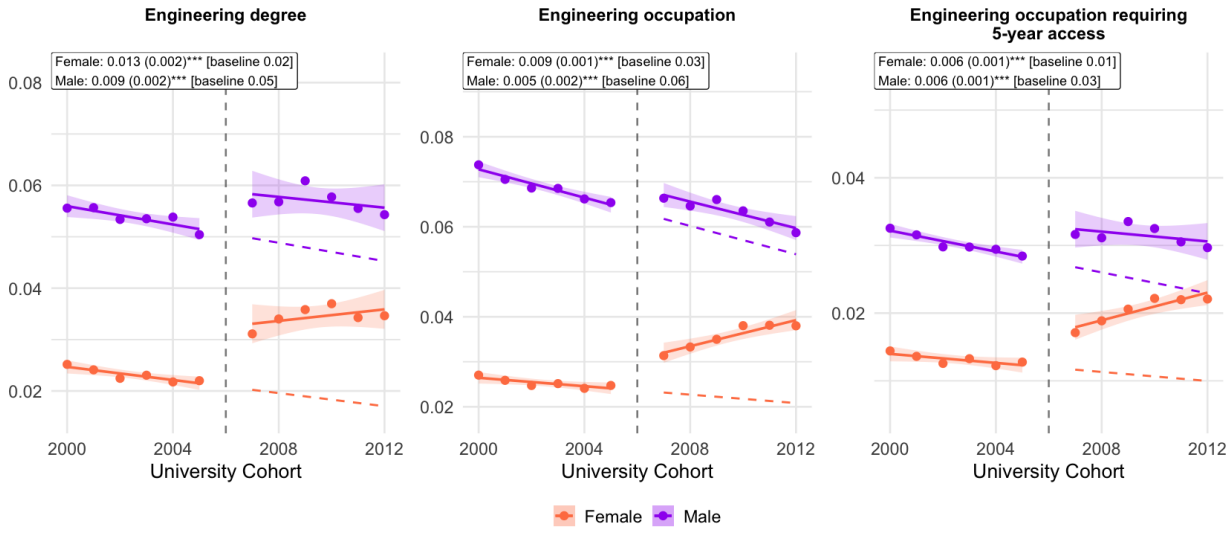


Figure 9: Engineering degrees and engineering occupations, by cohort

Notes: Quadros de Pessoal. Left: share of the cohort holding an engineering degree. Middle: share working in an engineering occupation. Right: share working in an engineering occupation that requires the full five-year cycle. See the notes to Figure 4.

of 4.2 percent, and employment in five-year engineering occupations by 0.6 percentage points on a base of 2.3 percent. Before the reform, 56 percent of engineering graduates worked in such occupations. If the new graduates took up these jobs at the same rate, the 0.6 point rise implies that about 1.1 percentage points more of each cohort completed the full five years—as many as the rise in engineering degrees itself. The reform brought students into engineering, and a fair share of them went on to the end. Outside STEM the reform shortened the first degree rather than adding an exit, and the question is whether the new graduates also continued to a master's. Of the additional non-STEM graduates, 17 percent hold a master's degree, almost the same as the 18 percent of pre-reform non-STEM graduates.[7] The marginal graduate in these fields is about as likely to continue as the average graduate before the reform.

Italy. The Italian survey separates degrees of four or more years from shorter ones, so it can check directly whether students stopped at three years. Table 3 shows the answer. Among women, engineering degrees rise by 0.4 percentage points, and engineering degrees of four or more years by 0.3 percentage points, so most of the new engineering graduates completed a long degree. Among men the engineering estimates are similar in size, 0.4 and 0.2 percentage points, but they are not statistically different from zero. Healthcare offers a contrast with Portugal. In Italy nursing was already a three-year diploma, and the reform turned it into a first degree with a master's on top. Nursing therefore did not become long relative to other fields, as it did in Portugal; instead it gained a route to further study. Healthcare degrees rise by 0.8 percentage points among women and 0.3 percentage points among men. Long healthcare degrees rise by 0.4 and 0.2 percentage points. In both countries, then, students moved towards the fields where the reform added a credential and away from those where it did not.

[7] Both shares come from *Quadros de Pessoal*, which codes the master's degree separately only from 2006 and uses a single *licenciatura* code for degrees of different lengths. The split between first-cycle and master's degrees is therefore less reliable than the effect on holding any degree (Section 3).

Table 3: Engineering and healthcare degrees by length, Italy

	Engineering		Healthcare	
	Any degree	4+ years	Any degree	4+ years
All	0.004***	0.002*	0.006***	0.003***
	(0.001)	(0.001)	(0.001)	(0.001)
Baseline	0.027	0.022	0.022	0.008
Women	0.004***	0.003**	0.008***	0.004***
	(0.001)	(0.001)	(0.002)	(0.001)
Baseline	0.012	0.009	0.030	0.009
Men	0.004	0.002	0.003*	0.002**
	(0.002)	(0.002)	(0.001)	(0.001)
Baseline	0.043	0.034	0.015	0.006
p-value, women = men	0.918	0.725	0.022	0.113

Notes: Italian Labour Force Survey, citizens aged 30–36. The running variable is the year a person turned 19 minus the adoption year in their region. RD estimates from individual-level regressions with separate linear trends on each side of the cutoff. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Baseline is the mean over pre-reform cohorts. "Any degree": any university degree in the field. "4+ years": a degree of four or more years (single-cycle or *laurea specialistica*) in the field. "All" pools women and men in one regression. Observations: 269,026 in all, 137,193 women and 131,833 men.

5.4 Where the wage gain comes from

Section 5.1 showed that post-reform cohorts earn 1.6 percent more at ages 28 to 30 (Table 1), and Section 5.2 that they work in different sectors and occupations. This subsection asks how much of the wage gain comes from where they work. I split it three ways: into the pay premium of the firm, by the type of firm, and into the job-title floor and the pay above it. Each split adds up exactly to the wage effect.

Better-paying firms. To see how much of the gain comes from where people work, I decompose the wage into a worker effect, a firm effect and year effects (Abowd et al., 1999), and estimate the RD on the firm effect of each worker's employer (Appendix C.6). Because the same decomposition applies to every worker, the RD on the wage equals the RD on the firm effect plus the RD on the rest. Column 2 of Table 4 shows that post-reform women work at firms that pay 0.9 percent more to all their workers, and men at firms that pay 0.4 percent more. Sorting into better-paying firms thus accounts for about two-fifths of the wage gain: 42 percent for women and 41 percent for men. The remaining three-fifths is the part of the wage that moves with the worker, such as their degree, field and occupation. This part is also different from zero, for women and for men. The firm share is the same for women and men, although women's gains are about twice as large in both parts.

Table 4: Hourly wages at ages 28–30: firm pay premium

	Log wage (1)	Firm effect (2)	Rest (3)
All	0.016***	0.007***	0.009***
	(0.003)	(0.001)	(0.002)
Women	0.022***	0.009***	0.013***
	(0.003)	(0.001)	(0.002)
Men	0.010***	0.004***	0.006**
	(0.003)	(0.001)	(0.003)
p-value, women = men	0.01	0.00	0.07
Share of the effect, all (%)	100	42	58

Notes: Quadros de Pessoal, workers aged 28–30 in the largest connected set, main job, cohorts 2000–2012 (2006 excluded; the 2012 cohort is observed at 28–29 only), observed in 2010–2023 (2.28 million worker-years). Log wage: log real hourly wage (monthly hours and earnings winsorised at the 99th percentile within year). Firm effect: $\hat{\psi}_{J(i,t)}$ from equation (10); rest = log wage minus firm effect. Column 1 = 2 + 3 exactly. RD on cohort effects net of age and calendar-year effects, estimated on cohort × age × year cells weighted by worker-years (the whole sample adds a sex dummy), linear on each side: 6 pre- and 6 post-reform cohorts. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Which firms. Which firms are these? Because the reform changes who works where, comparing wages within a type of firm would compare different people. Instead, for every worker I take the log wage minus its pre-reform mean if the worker is employed in a given type of firm, and zero otherwise, and estimate the same RD on it (Appendix C.7). The four types in Table 5 cover every worker, so the four RDs add up exactly to the wage effect. IT and engineering firms account for 16 percent of the effect, business and support services firms for 12 percent, and banking, insurance and finance for 21 percent. These firms employ a fifth of pre-reform workers but account for half of the effect. They are the sectors that post-reform cohorts moved into (Section 5.2). The banking share should be read with care. It reflects a decline in banking employment among pre-reform cohorts that stops after the cutoff, rather than a break at 2007.

Job-title floor and pay above it. A third split asks how the gain is paid, following the approach in Cardoso and Portugal (2005) explained in Appendix C.8. Most Portuguese private-sector workers are covered by collective agreements that set a minimum rate for each job title. A higher wage can therefore come from a higher floor, the rate attached to the worker's job title, or from a larger cushion, the pay a firm adds above that rate. I measure the floor of a job title in a year as the modal hourly wage of the workers who hold it, and split each worker's wage into the floor and the cushion. Eighty-one percent of workers aged 28 to 30 hold a job title with a measured floor, and on this sample the wage gain is 1.9 percent (Table 6). The cushion accounts for 1.1 percentage points of it and the floor for 0.8. Both are different from zero. Women and men differ. For women the cushion is the larger part, 1.8 of 3.0 percent. For men the two parts are the

Table 5: Hourly wages at ages 28–30: where the effect accrues, by the type of firm people work in

		Part of the effect in			
	Log wage (1)	IT and engineering (2)	Business and support services (3)	Banking, insurance and finance (4)	All other firms (5)
All	0.016***	0.003***	0.002***	0.003***	0.008**
	(0.003)	(0.000)	(0.000)	(0.001)	(0.003)
Women	0.022***	0.003***	0.003**	0.004***	0.012***
	(0.003)	(0.001)	(0.001)	(0.001)	(0.003)
Men	0.010***	0.003**	0.001	0.003***	0.004
	(0.003)	(0.001)	(0.001)	(0.001)	(0.003)
p-value, women = men	0.01	0.90	0.06	0.24	0.11
Share of the effect, all (%)	100	16	12	21	51
Share of workers, pre-reform (%)	100	4.2	13.9	2.5	79.4

Notes: Quadros de Pessoal, sample and log wage as in Table 4. Column 1 is the RD on the log wage. Columns 2–5 are RDs on (log wage minus its pre-reform mean) times an indicator for working in that type of firm, zero otherwise, defined for every worker; they add up exactly to column 1. No column conditions on the type of firm. A type of firm adds to the effect when more people work there at above-average pay, or when those who work there are paid more. IT and engineering: CAE divisions 62, 63 and 71. Business and support services: 701, 702, 692, 749 and divisions 77–82. Banking, insurance and finance: divisions 64–66. RD on cohort effects net of age and calendar-year effects, estimated on cohort × age × year cells weighted by worker-years (the whole sample adds a sex dummy), linear on each side: 6 pre- and 6 post-reform cohorts. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

same size, 0.5 percentage points each; the floor is different from zero only at the 10 percent level, and the cushion is not.

Table 6: Hourly wages at ages 28–30: job-title floor and pay above the floor

	Log wage (1)	Floor (2)	Cushion (3)
All	0.019***	0.008***	0.011**
	(0.003)	(0.002)	(0.003)
Women	0.030***	0.011***	0.018***
	(0.004)	(0.003)	(0.003)
Men	0.010***	0.005*	0.005
	(0.003)	(0.003)	(0.004)
p-value, women = men	0.00	0.14	0.01
Share of the effect, all (%)	100	42	58

Notes: Quadros de Pessoal, sample and log wage as in Table 4, restricted to workers whose job title has a floor (81 percent of worker-years). Floor: modal hourly wage of the job title (*categoria profissional*) in the year, cells of at least ten worker-years; cushion = log wage minus floor. Column 1 = 2 + 3 exactly. RD on cohort effects net of age and calendar-year effects, estimated on cohort × age × year cells weighted by worker-years (the whole sample adds a sex dummy), linear on each side: 6 pre- and 6 post-reform cohorts. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Taken together, the results describe a reform that raised the number of graduates, moved students towards the fields where an earlier credential appeared, and raised rather than reduced the number completing the full degree. Pay rose, and about two-fifths of the rise came from working at better-paying firms. Section 6 develops a model in which students learn about their match with a field while enrolled, estimates it on the education effects, and uses it to predict the effect of the reform on wages without targeting it.

6 A model of trying university [work in progress]

The results show that students moved towards both kinds of field the reform restructured, that more of them completed university, and that more of them reached engineering jobs that require the five-year degree. Section 1 offered an interpretation. Where the first degree became shorter, the reform lowered its cost. Where an intermediate credential appeared inside a five-year degree, the reform lowered the risk of trying it. This section writes down the simplest model in which both channels operate, assuming that employers absorb any change in the number of graduates at unchanged wages. I estimate it on the education effects alone and use it to answer two questions. Which channel drives each response? Appendix I gives the details.

6.1 The model

Fields. The model has three fields, defined by what the reform did to them, as in Section 5.2. In the first, the reform introduced an intermediate credential: a five-year degree became a three-year degree followed by a two-year master's. In the second, it shortened the first degree from four years to three. In the third, nothing changed: the degree still takes four years and has no earlier exit.

Students and beliefs. A student's match with each field is either good or poor. She does not know it. She holds a belief p_f that her match with field f is good, and this belief is correct on average. Her starting belief is

$$\operatorname{logit} p_{if} = \operatorname{logit} \pi_i + a_f + s\,\varepsilon_{if}, \qquad \varepsilon_{if} \sim N(0,1), \tag{3}$$

where π_i is her predicted probability of expecting to complete university from PISA 2003 for Portugal. The term ε_{if} is what she knows about her fit with field f beyond what PISA predicts, and s measures how much this varies across students. The field shifts a_f are set so that the model matches the pre-reform share of each cohort holding a degree in each field. Only the field shifts and each student's private information ε_{if} differ across fields, and ε_{if} is drawn independently for each field. The PISA prior, the spread s and the speed of learning described below are the same in every field.

Progress and learning. A good match passes every year. A poor match fails a year with probability q; a failed year earns no credit and must be repeated. A student therefore learns about her match from her

own progress: a failed year shows that she is a poor match, and each year passed makes a good match more likely. At the end of her third year she also receives a signal that reveals her match almost fully.

Credentials and wages. Each credential pays its *Quadros de Pessoal* wage relative to workers without university: 1.59 for the intermediate credential, 1.67 for the shorter first degree and 1.65 in the unchanged field. The full five-year degree pays 1.83 to a good match. A poor match earns only the wage of the intermediate credential with it, so the two extra years pay off only for good matches. Without a credential everyone earns the wage of workers without university.

Payoffs. A student who leaves after t years with yearly wage w has the lifetime payoff

$$W(t,w) = \frac{-k\sum_{j<t}\beta^j + w\sum_{j=t}^{R-1}\beta^j}{\sum_{j=0}^{R-1}\beta^j}, \tag{4}$$

expressed as a constant yearly income for life, so that working at once gives $W = 1$. Each year of study costs her the forgone wage and a direct cost k, and working life ends R years after entry. Here j counts the years since entry: the first sum runs over her years of study and the second over her years of work. She values her lifetime income with a dislike of risk: every choice maximises the expected value of $u(W) = (W^{1-\gamma}-1)/(1-\gamma)$, with $\gamma = 6$.

The student's problem. A programme in field f under regime r is a pair (L_{fr}, T_{fr}): the intermediate credential needs L_{fr} years of credit and the full degree T_{fr}, with $L_{fr} = \infty$ where there is no intermediate credential. Let $V_{fr}(t,c,b)$ be the expected utility of a student at the end of year t with c years of credit and belief b. She chooses to leave with nothing, to leave with the intermediate credential if she has reached it, or to continue:

$$V_{fr}(t,c,b) = \max\Big\{u\big(W(t,1)\big),\ \mathbf{1}[c \ge L_{fr}]\,u\big(W(t,w_f^L)\big),\ C_{fr}(t,c,b)\Big\}. \tag{5}$$

Continuing means studying one more year. With probability $(1-b)q$ she fails, keeps c credits and learns that she is a poor match; otherwise she passes and gains a credit:

$$C_{fr}(t,c,b) = (1-b)q\,V_{fr}(t+1,c,0) + \big[1-(1-b)q\big]\,\mathrm{E}\,V_{fr}(t+1,c+1,b^+), \tag{6}$$

where $b^+ = b/[b+(1-b)(1-q)]$ is her belief after passing, further updated by the signal at the end of the third year, over which the expectation is taken. When her credits reach T_{fr} she graduates, with expected

utility $b\,u(W(t, w_f^T)) + (1-b)\,u(W(t, w_f^L))$; a poor match earns w_f^L, which equals w_f^T in the fields without an intermediate credential.

Entry. The value of entering field f is $C_{fr}(0, 0, p_{if})$, and its certainty equivalent $\mathrm{CE}_f^r(p_{if})$ is the constant yearly income for life that she values as much. Before the first year she takes the best of working and the three fields:

$$d_i^r = \arg\max \left\{1,\ \mathrm{CE}_1^r(p_{i1}),\ \mathrm{CE}_2^r(p_{i2}),\ \mathrm{CE}_3^r(p_{i3})\right\}. \tag{7}$$

There are no tastes for fields and no random shocks at entry: a student's field reflects only her beliefs and what each field pays. Fields are therefore substitutes, and a shorter first degree draws students from the other fields as well as from work.

Where the reform acts. The two changes work differently. Where the first degree became shorter, it arrives a year earlier. This saves a year of costs and forgone earnings for every student who graduates, so it is worth most to students who expect to finish. Where an intermediate credential appeared, every plan available before the reform is still available after it, and students learn exactly as before. What changes is the bad outcome. Before the reform, a student who found that she was a poor match could leave with nothing or stay five years for a degree that paid her no more than three would have. After it, she can leave after three years with a credential. The intermediate credential therefore raises the value of trying the field without changing the value of succeeding. This is the sense in which the reform leaves students *less to lose*.

6.2 Estimation and fit

I fix the discount factor, the working life, the direct cost, the signal at the end of the third year, the wages and the dislike of risk. I estimate two parameters by minimum distance on the five education moments in Panel A of Table 7: the spread of beliefs s and the poor match's chance of failing a year q. The field shifts a_f are re-set at every evaluation. The two field moments are the effects in Figure 6, averaged over women and men. The five-year moment counts good matches who complete the five-year degree, because the data measure employment in engineering jobs that require it. The estimates are $s = 4.3$ and $q = 0.54$: a poor match fails about one year in two. Wages are not targeted.

Table 7: The model and the data

Moment	Data	(SE)	Model	Source
A. Targeted				
Δ share with a degree that gained an intermediate credential (pp)	1.70	(0.21)	1.62	QP RD
Δ share with a degree whose first cycle became shorter (pp)	1.62	(0.15)	1.67	QP RD
Δ share in engineering jobs needing five years (pp)	≥0.59	(0.11)	0.97	QP RD
Entrants leaving with no degree, before (%)	17	(4)	16.4	IEFA
Δ share of the cohort leaving with no degree (pp)	2.6	(2.9)	1.59	IEFA RD
B. Not targeted				
Δ log hourly wage, ages 28–30 (log points)	1.6	(0.3)	1.8	QP RD
Δ share completing university (pp)	2.2	(0.4)	3.0	QP RD
Δ share with a degree in an unchanged field (pp)	−1.63	(0.12)	−0.28	QP RD

Notes: RD effects at the 2007 cutoff, cohorts 2000–2012, 2006 excluded; shares of the cohort in percentage points. Field shares: share of the cohort holding a degree in each group of fields of Figure 6, average of the estimates for women and men (standard error of the average). The five-year moment is one-sided: the model must reach at least the data value. Dropout: entrants of cohorts 2000–2005 and 2007–2012 in the 2016 and 2022 IEFA waves, ages 22 and above, with the corrected coding of leaving university. Wage: Table 1, column 1; in the model, the change in the cohort's mean wage once everyone has left university. Completion: QP, share whose highest level is a university degree, average of the estimates for women (2.9, SE 0.6) and men (1.4, SE 0.4) (Appendix Table 13). Source: `outputs/model/simple/MIN_paper_fit.txt`.

The model reproduces every targeted response within half a standard error. Estimated on education moments alone and assuming employers absorb the additional graduates at the same wage, the model predicts a wage gain of 1.8 percent, against 1.6 percent in the data. Of the 1.8 percent, 0.96 comes from more graduates with a shorter first degree, 0.67 from more students completing the five-year degree and 0.35 from students leaving with the new intermediate credential; fewer graduates in the unchanged fields subtract 0.15 (Appendix I.5).

6.3 Cost and risk

Figure 10(a) shows what the reform is worth to a student, by her starting belief in the field. The gain counts her option not to enter: it is zero for a student who would not enter the field before or after the reform. The unchanged field's line is zero throughout. In both reformed fields, no student with a belief below 0.35 gains, because she would not enter even after the reform. Between 0.35 and the pre-reform entry threshold (0.54 for the intermediate credential, 0.49 for the shorter first degree), the gain rises steeply: these are students for whom the reform makes entering worth more than working. Above the threshold, the two curves move in opposite directions. The gain from the shorter first degree keeps rising, to 8.5 percent of a yearly wage, every year for life, for a student who is sure of a good match. It is a cost saving, and only students who

graduate collect it. The gain from the intermediate credential peaks at 4.7 percent just above the threshold and falls to almost nothing for a student who is sure of a good match. It is insurance: it pays in the outcome the student fears. On average, the intermediate credential is worth 2.7 percent of a yearly wage to a student who would have entered its field anyway, and the saved year 6.1 percent.

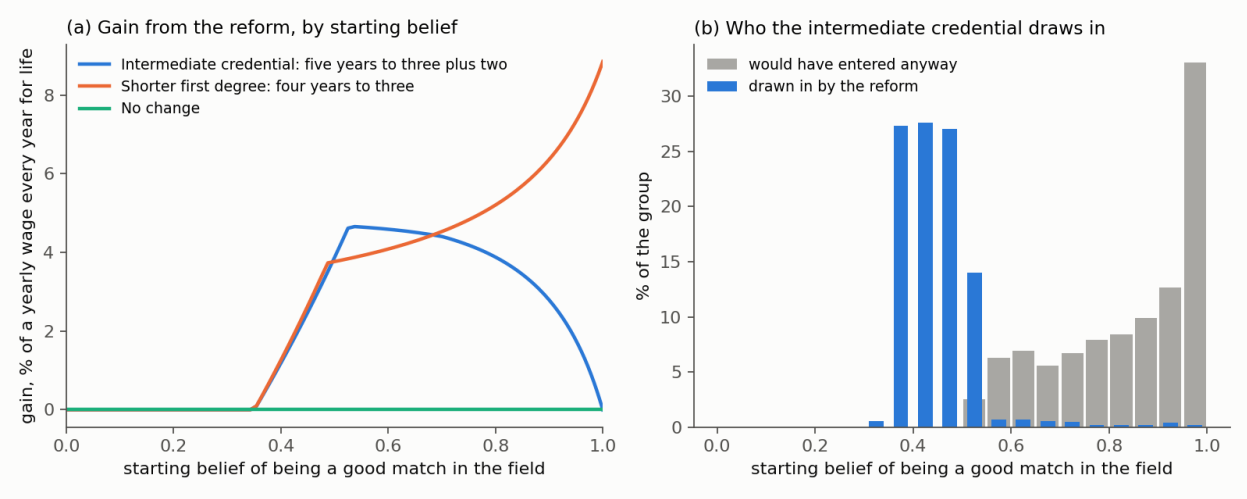


Figure 10: Value of the reform and who the intermediate credential draws in

Notes: (a) Gain from the reform to a student with a given starting belief in the field, counting her option to work instead: $\max\{\text{CE}_f^{\text{post}}(p), 1\} - \max\{\text{CE}_f^{\text{pre}}(p), 1\}$, in percent of a yearly wage for every year of working life. Each curve compares the field with working only, so it ignores the other fields. (b) Starting beliefs of students who would have entered the field with the intermediate credential anyway and of students the reform draws into it. Model at the estimates in Section 6.2.

Because the intermediate credential is worth most to unsure students, it draws in students who would not otherwise have tried the field (Figure 10(b)). Their average starting belief is 0.45, against 0.84 for students who would have entered anyway. New entrants account for most of the rise in degrees in this field: the reform draws 2.7 percent of the cohort into it, while among students who would have entered anyway, the share leaving with nothing falls by only 0.3 points. Some of the new entrants turn out to be good matches and continue. Of every 100 new entrants, 44 complete the five-year degree, 12 leave with the intermediate credential and 44 leave with nothing. Few students who would have completed five years stop at three instead (0.06 percent of the cohort), so the number of five-year graduates rises, by 0.9 points.

Table 8 separates the two channels. It holds every parameter at its estimate, changes one feature of the world, and reports the reform's effect in that world. Two results stand out.

First, each change works through its own field. The intermediate credential alone raises degrees in its field by 2.0 percentage points. The shorter first degree alone raises degrees in its field by 1.8 points. Each

lowers degrees in the other fields slightly.

Second, uncertainty matters for both changes, but in different ways. The last row lets every student learn her match at the end of her first year, before the intermediate credential can matter. The intermediate credential is then worth nothing. It draws no one in, degrees in its field no longer rise, and the rise in good matches completing five years disappears. The saved year keeps two-thirds of its value, and degrees with a shorter first degree still rise, by 0.6 points, a third of the effect. The shorter first degree attracted students by lowering cost, and its value grows when students are unsure whether they will progress. The intermediate credential attracted them only by lowering risk.

Table 8: The reform's effect with one feature changed

	Degrees, by what the reform did			University	Five-year	Value (% of a yearly wage)	
	Intermediate credential	Shorter first degree	No change	entry	degree	Intermediate credential	Saved year
The reform	1.62	1.67	−0.28	4.60	0.97	2.72	6.13
Intermediate credential only	1.95	−0.11	−0.09	2.59	1.28	2.72	–
Shorter first degree only	−0.28	1.79	−0.18	2.14	−0.28	–	6.13
Match learned after year 1	−0.25	0.55	−0.13	0.49	−0.27	0.01	4.03

Notes: Changes in percentage points of the cohort. Five-year degree: good matches completing the five-year degree. Value: average gain to students who would have entered the field before the reform, every year for life. Each row holds the estimated parameters and field shifts fixed, changes one feature, and compares the pre- and post-reform menus within that world. In the last row, a signal at the end of the first year reveals the match. Further rows and outcomes: Appendix Table 21. Source: `outputs/model/simple/MIN_paper_cf.txt`.

Limits. The model is deliberately simple, and three of its features deserve caution. First, the dislike of risk is fixed, and the model needs it. Refitted without it, the model cannot produce the rise in five-year completion. Switched off at the estimates, it makes entry respond more than ten times as strongly, so it cannot be used to isolate a channel (Appendix I.3). Second, beliefs are widely spread: PISA accounts for about an eighth of their variation, and a third of students who would have entered the field with the intermediate credential anyway are more than 95 percent sure of a good match. Third, students learn their match quickly and fully: almost every student leaves university nearly sure of her match, the model has no master's after the shorter first degree.

7 Conclusion

When Europe launched the Bologna Process, advocates expected shorter and more flexible degrees to widen access to university. Critics expected students to spend less time in university and to earn less as a result. The evidence from Portugal and Italy supports the advocates' hope more than the critics' worry. More students graduated. On net, they spent more years in university, not fewer, and by their late twenties they earned more.

The reform used two different tools, and they serve different purposes. Shortening a first degree is a cost policy. It saves a year for every student who finishes, so it helps most the students who already expect to finish. An intermediate qualification works like insurance. It changes nothing for a student who completes the full degree. It pays off only when a student finds that the field does not suit her: she can then leave with a qualification instead of nothing. It is therefore worth most to students who are unsure, and these are the students it draws in. A country can lower the risk of entering a long programme without making the programme any shorter.

Two conditions made this work in Portugal. First, the full degree kept its value. The title of engineer still required all five years, so students who turned out to be well matched had a reason to continue, and many did. How the old and new degrees are recognised matters. In Italy, the old four-year degree was equated with the new master's, so a qualification of the old standing took longer to reach after the reform, and the response in those fields was smaller. Second, employers absorbed the extra graduates. They worked in jobs that matched their field at the same rate as earlier graduates, and much of their wage gain came from working at higher-paying firms. Where employers cannot absorb more graduates in a field, the same reform may raise attainment without raising pay.

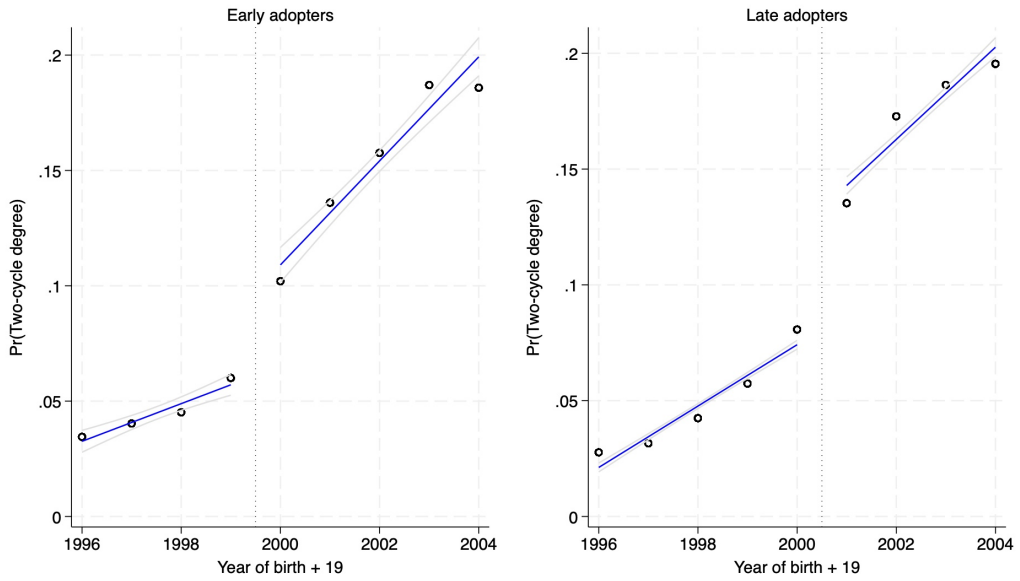
These lessons reach beyond Europe. In Latin America, first degrees still take four to five years, and countries such as Argentina and Chile have recently shortened their degrees or are reviewing their length. As in Europe two decades ago, the debate there is mostly about how many years a degree should take. This paper suggests that a second question matters as much: what students leave with if they stop before the end. A long programme with a recognised exit partway through can widen access and still keep the training of the full degree.

A Institutional Setting: Details

A.1 Timing of adoption

This subsection adds detail to Figure 1. In Portugal, Decree-Law 74/2006 (March 2006) let programmes convert from 2006/07 and required conversion by 2008/09. In Italy, Ministerial Decree 509/1999 took effect once the first-level degree classes were published in October 2000, with 18 months to convert. In practice, Bondonio and Berton argue that the timing of adoption was close to random, because it depended on when each faculty council first met after the decree (Bondonio and Berton, 2018). The Italian design moves the cutoff to 2000 in regions where universities adopted early, as shown in figure 11.

Figure 11: Staggered adoption of the reform in Italy



Notes: Italian Labour Force Survey, 2009–2020. Share of respondents whose highest degree is a two-cycle degree, by the year they turned 19, with linear fits on each side of the adoption year. Early adopters: Piemonte, Valle d'Aosta, Friuli-Venezia Giulia and Sardinia, where universities adopted the reform in 2000/01; late adopters: all other regions (2001/02). Regions are assigned by region of residence.

A.2 What the three-year degree is worth

An earlier credential lowers the cost of a failed attempt only if employers accept it. In Portugal they were in effect obliged to, because the new degree kept the old name. The law never distinguishes licenciaturas by length or date: entry to the graduate career of the civil service requires "a licenciatura or higher degree"

(Law 12-A/2008, art. 44), and sectoral collective agreements tie wage floors to the terms "licenciado" and "bacharel" without mentioning Bologna. By default a three-year licenciado is therefore placed on the grade a four- or five-year licenciado used to occupy. Attempts to separate the two failed: the Constitutional Court struck down the bar association's entrance exam for post-Bologna law graduates (Acórdão 3/2011), and labour courts ruled against an employer that hired only pre-Bologna licenciados. In regulated professions the value of the exit differs by field. Three-year engineering graduates can register as technical engineers, the title previously held by holders of the three-year bacharelato, and accountancy is open to any suitable licenciatura; the three-year degree carries no licence in medicine, psychology or school teaching, where a master's became compulsory.

Italy gave the new degree a lower but still defined place. Decree 328/2001 split the professional registers of engineers, architects, biologists, chemists, geologists and psychologists into a section for master's graduates and a "junior" section open to three-year graduates, and the three-year degree gives access to graduate grades in local government. Unlike in Portugal, however, the old four-to-six-year laurea was formally equated with the new master's, not with the three-year degree, for public-sector competitions. The pooling of old and new credentials that operates in Portugal is therefore weaker in Italy.

A.3 Other changes at the same time

Conversion of enrolled students. In both countries students already enrolled could convert to the new degrees, with credit for the exams they had passed. Cohorts just before each cutoff could therefore still leave with the new credential, so estimates for completion and later outcomes are, if anything, understated.

Entry caps and nursing (Italy). Law 264/1999 capped entry nationally to medicine, veterinary medicine, dentistry, architecture, the health professions and primary-teacher education, so entry in those fields reflects the number of places set each year. Nursing entry grew throughout 1997–2010 as training moved from regional schools into universities; the reform renamed the nursing diploma a first degree but did not shorten it.

Fees. Tuition fees did not change at either cutoff.

B Data: Details

B.1 Variable definitions

Educational attainment.—The main education outcome is an indicator for holding a university degree as the highest completed level. In QP this covers the *bacharelato*, *licenciatura*, *mestrado* and doctorate; once a worker is observed with a degree, the indicator stays on in later years. In the Portuguese surveys it covers the corresponding higher-education codes, and in Italy any tertiary title. In Portugal, the split between *licenciatura* and *mestrado* is only available in the data for some years, hence I refrain from using this distinction and use year of education proxies where possible. In Italy the corresponding variables are a two-cycle degree, which measures take-up of the new structure, and a degree of four or more years (a single-cycle degree or a *laurea specialistica*).

Years of university.—In the surveys, years of university equal the age at which the respondent left education minus 18 (19 in Italy) for graduates, and zero for everyone else. QP has no such question. I proxy it with the age at which a worker is first observed with their highest degree, minus 18; because this also counts any delay before the first private-sector job, I use it only as a proxy, validated against the surveys.

Field of study.—I group fields by what the reform did to the length of the first credential (Section 2). In QP the field comes from the last two digits of the education code, which allows me to separate between STEM, where a five-year programme gained a three-year exit; economics and management, the social sciences and the humanities, where a four-year programme became three plus two; and fields with no change, namely health and law. The Portuguese surveys use the broad fields of the national classification (CNAEF), in which law cannot be separated from the social sciences and business. In Italy I group degree subjects into STEM, humanities and teaching, the social sciences (including law and psychology), and health.

Occupation and job level.—I classify occupations into managers, professionals, technicians, clerical workers, craft workers and low-skill occupations (service and sales workers, operators and elementary occupations). QP switches from the CNP-94 to the CPP-2010 classification in 2010. I map each CNP-94 code to its most common CPP-2010 major group using INE's correspondence table, which matches 98.9 percent of worker-years. Separately, I use the worker's hierarchical level within the firm, whose top two levels are senior

and middle management (*quadros*). This is a job level, not an occupation. In Italy occupations follow the CP2011 major groups.

Wages and earnings.— The QP hourly wage is monthly October pay divided by normal monthly hours, deflated to 2010 euros with the consumer price index. Earnings and hours are each capped at their 99th percentile within year before the ratio is taken. Wages need one further step, because post-reform cohorts reach a given age in later calendar years. Pre-reform cohorts reach their late twenties during the 2011–14 crisis, and post-reform cohorts during the recovery and the minimum-wage increases after 2015. The baseline wage therefore pools ages 28 to 30. In this window each calendar year contains several cohorts at different ages, so I include calendar-year effects alongside age effects, estimated from the workers in the sample. A common linear trend cannot be split between cohorts and years, but it does not change the jump at the cutoff. At a single age this is not possible, because the year is then an exact linear function of the running variable. For wages at a single age I instead subtract a calendar index estimated outside the cohort panel. The index is the mean log real hourly wage of workers aged 40–45 in each year, computed within cells of sex, education and NUTS2 region and weighted by each cell's 2005 share of workers, so it moves with the wage level and not with who is employed. It is normalised to zero in 2005.

I measure cumulative real earnings to a given age by summing annual earnings, set to zero in years the worker is not in QP. I pad each worker's record to one row per age from school-leaving age to age A, set annual earnings to zero where no earnings are observed, and sum. Zeros capture delayed entry; the measure does not condition on employment.

To separate pay set by the job from pay set by the employer, I define the floor of a job title as the most common log hourly wage within job title and year (in cells of at least ten workers) and the cushion as the gap between a worker's log wage and that floor. In the surveys employment follows the ILO definition and pay is set to zero for those not employed.

Enrolment and dropout.—From IEFA I define entry to higher education as having completed a degree or having started one and left, and dropout as having started and left without a degree. I report dropout as a share of the whole cohort.

C Empirical Strategy: Details

C.1 What the estimates measure

The coefficient α_1 measures the effect of facing the reformed degree structure at the age of entry, averaged over the whole cohort. The average includes people who never enrol, because they are part of the population whose choices the reform could change. I do not rescale α_1 by a first stage. The reform did not offer a treatment that some students took up and others refused; it changed the options open to every student, both the decision to enrol and the decision of when to stop.

C.2 Estimation and standard errors

I estimate equation (2) on cohort means, in two steps. For the education outcomes, I first remove a quadratic in age from each worker-year and average the outcome within each cohort, separately for women and men. For wages, the first step instead regresses mean wages in cohort $\times$ age $\times$ calendar-year cells on cohort, age and year dummies, weighting each cell by its number of worker-years; the cohort coefficients are the cohort means (Appendix C.4). In the second step, I fit a line on each side of the cutoff to the twelve cohort means by ordinary least squares. The standard errors are the conventional standard errors of this second regression, and p-values come from a t distribution with eight degrees of freedom (twelve cohorts minus four parameters). The cohort is therefore the unit of inference. This follows Donald and Lang (2007): with few groups, cluster-robust standard errors computed on individual observations are biased downwards, whereas a regression on group means, with a t distribution whose degrees of freedom are the number of groups minus the number of parameters, gives valid inference when the cohort-level errors are close to normal. This allows for shocks common to everyone in a cohort, such as the labour market they entered, which standard errors computed on individual worker-years would ignore. Each cohort mean rests on more than a hundred thousand worker-years, so the sampling error of the first step is small next to the variation across cohorts, and I do not add it. To test whether the effect differs between women and men, I compare the two estimates with a z-test that treats the two samples as independent. The Italian regressions are run on individual respondents, with heteroskedasticity-robust standard errors.

C.3 Validity of the design

Density of the running variable Birth year cannot be manipulated, but the number of workers per cohort in *Quadros de Pessoal* could still jump at the cutoff if the reform changed who enters the private sector. I test for a break in the density of the running variable in the spirit of McCrary (2008). For each cohort bin j I compute $\hat{f}(z_j) = N_j/(Nh)$, where N_j is the number of workers in the bin, N the total and $h = 1$ the bin width, and estimate

$$\log \hat{f}(z_j) = \gamma_L + \delta_L z_j + u_j \quad (z_j < 0), \qquad \log \hat{f}(z_j) = \gamma_R + \delta_R z_j + v_j \quad (z_j \geq 0), \tag{8}$$

weighting each bin by N_j and using a triangular kernel within a symmetric window around the cutoff. The discontinuity in the log density is $\hat{\gamma}_R - \hat{\gamma}_L$. Figure 12 shows no visible break. The estimated jump in the log density is 0.019, with a standard error of 0.020.

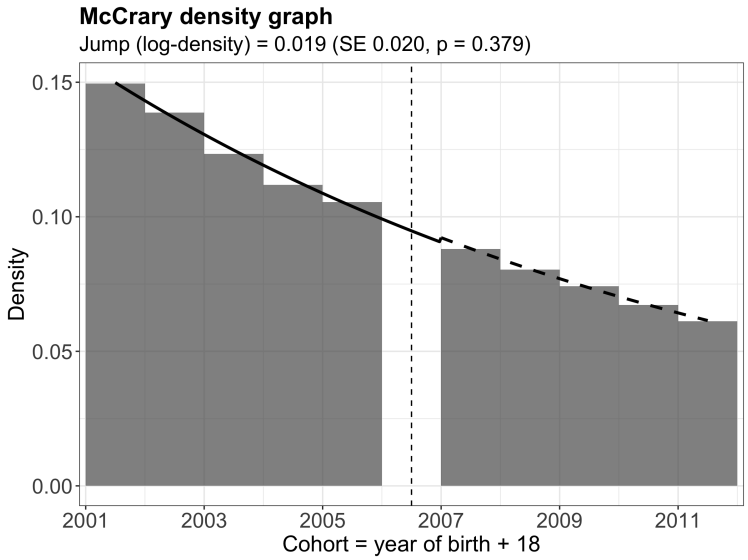


Figure 12: Density of the running variable in *Quadros de Pessoal*

Notes: Quadros de Pessoal. Share of workers by the year they turned 18, with fits from equation (8) on each side of the cutoff; the dotted line marks the excluded 2006 cohort. The estimate at the top of the panel is the jump in the log density at the cutoff.

No delayed entry Students who turned 18 shortly before 2007, and who expected the reform, could have postponed entry to enrol in the shorter degrees. Such students would then be counted as untreated although they studied under the new structure, and the pre-reform cohorts closest to the cutoff would be selected. Postponing entry usually means working in the meantime, so I estimate equation (2) with an indicator for

being observed in employment between leaving secondary school and completing a university degree. Table 9 shows that the discontinuity is close to zero and precisely estimated for women and men alike.

Table 9: Employment between secondary school and university

	Whole sample	Women	Men
RD jump	0.001	0.002	−0.000
	(0.001)	(0.002)	(0.002)
Baseline	0.105	0.128	0.085
p-value, women = men		0.283	

Notes: Quadros de Pessoal. The outcome is an indicator for being observed in employment between leaving secondary school and completing a university degree. Estimates of equation (2). Standard errors in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

C.4 How the wage effect at ages 28–30 is estimated

Each worker-year in *Quadros de Pessoal* has a cohort c (the year the worker turned 18), an age a and a calendar year t, with $c = t - a + 18$. I keep worker-years at ages 28 to 30 and compute the mean log hourly wage $\bar{w}_{cat}$ in each cohort × age × year cell. The estimate then takes two steps. First, I regress the cell means on a full set of cohort, age and calendar-year dummies, weighting each cell by its number of worker-years:

$$\bar{w}_{cat} = \beta_c + \lambda_a + \mu_t + e_{cat}. \tag{9}$$

The cohort coefficients $\hat{\beta}_c$ are the cohort points. Second, I fit a line on each side of the cutoff through these points, $\hat{\beta}_c = \alpha_0 + \alpha_1 D_c + \alpha_2 Z_c + \alpha_3 D_c Z_c + u_c$. The jump $\hat{\alpha}_1$ is the estimate in Table 1, column 1.

The age effects λ_a remove the rise of pay with age. Without them, the 2012 cohort, observed only at ages 28 and 29, would look poorer than the others. The year effects μ_t remove changes in pay that hit everyone in a given year, such as the crisis, the recovery or a rise in the minimum wage. With a fixed age window, each cohort is observed in different calendar years. Without year effects, the cohort points would pick up the business cycle.

Cohort, age and year are tied: knowing two of them gives the third. The regression therefore cannot tell a steady trend across cohorts from a steady trend across years. This does not affect the estimate. Adding the same trend to every cohort point tilts both lines by the same amount, so the jump at the cutoff and the change in slope stay the same. The slope on either side, by contrast, has no meaning on its own.

Table 10 shows which cohorts are observed in each year. Within a year, the cohorts present differ by age. Pre- and post-reform cohorts appear in the same year only once: in 2017, the 2005 cohort is 30 and the 2007 cohort is 28 (the 2006 cohort is dropped). In every other year, all the cohorts present are on the same side of the cutoff. Those years tell the regression how much more 30-year-olds usually earn than 28-year-olds. The jump then comes from 2017. It measures how much more the post-reform 28-year-olds earn, relative to the pre-reform 30-year-olds, than 28-year-olds usually earn relative to 30-year-olds. The assumption is that, without the reform, the pay gap between ages 28 and 30 would have been the same in 2017 as in other years. Put differently, a shock in a given year moves the pay of 28-, 29- and 30-year-olds by the same amount.

Three checks bear on this assumption. First, wider age windows make the comparison in more years: pre- and post-reform cohorts meet in 2017 to 2019 at ages 28 to 32, and in 2016 to 2020 at ages 27 to 33. Both give the same effect (Table 11, columns 2 and 3). Second, dropping 2017 from these wider windows leaves the effect unchanged: 2.2 percent for women in both, and 1.3 and 1.2 percent for men. The estimate therefore does not rest on one year. Third, the placebo cutoffs in Appendix D make the same kind of comparison at cohorts where nothing happened.

Table 10: Cohorts observed at ages 28–30, by calendar year

Calendar year	Age 28	Age 29	Age 30
2010	2000		
2011	2001	2000	
2012	2002	2001	2000
2013	2003	2002	2001
2014	2004	2003	2002
2015	2005	2004	2003
2016	–	2005	2004
2017	**2007**	–	2005
2018	**2008**	**2007**	–
2019	**2009**	**2008**	**2007**
2020	**2010**	**2009**	**2008**
2021	**2011**	**2010**	**2009**
2022	**2012**	**2011**	**2010**
2023		**2012**	**2011**

Notes: Each entry is the cohort (year of turning 18) observed at that age in that year, $c = t - a + 18$. Post-reform cohorts in bold. "–" marks the excluded 2006 cohort. Blank cells fall outside cohorts 2000–2012. The year 2017 is the only one with both pre- and post-reform cohorts.

C.5 Wage effects at other ages and with the ages 40–45 index

Table 1 measures the wage at ages 28 to 30 and removes age and calendar-year effects. The effect does not depend on the ages used. Widening the window to ages 28 to 32 or 27 to 33 gives 1.6 to 1.7 percent (Table 11, columns 2 and 3), with the same gap between women and men. Columns 4 and 5 remove calendar time differently: they subtract from each wage the mean log wage of workers aged 40 to 45 in the same year. This is the only option at a single age, because cohort and calendar year then move together. It gives larger effects, 3.0 percent at age 30 and 3.4 percent at ages 28 to 30. The gap comes mainly from 2014 to 2016, at the end of the crisis, when the wages of young graduates fell further than those of older workers. Calendar-year effects estimated on the workers in the sample remove this, so I use them in the baseline.

Table 11: Effect of the reform on log hourly wages: other ages and the ages 40–45 index

	Age and calendar-year effects			Ages 40–45 index	
	Ages 28–30 (1)	Ages 28–32 (2)	Ages 27–33 (3)	Age 30 (4)	Ages 28–30 (5)
All	0.016***	0.017***	0.016***	0.030**	0.034***
	(0.003)	(0.003)	(0.003)	(0.010)	(0.005)
Women	0.022***	0.022***	0.022***	0.035**	0.038***
	(0.003)	(0.004)	(0.004)	(0.010)	(0.005)
Men	0.010***	0.012***	0.012***	0.023**	0.029***
	(0.003)	(0.002)	(0.002)	(0.010)	(0.005)
p-value, women = men	0.01	0.02	0.03	0.43	0.24
Cohorts, pre / post	6 / 6	6 / 6	6 / 6	6 / 5	6 / 6
Calendar years	2010–2023	2010–2023	2009–2023	2012–2023	2010–2023
Worker-years (millions)	2.28	3.56	4.86	0.73	2.28

Notes: *Quadros de Pessoal*, main job, workers in the largest connected set. Outcome: log real hourly wage (monthly hours and earnings winsorised at the 99th percentile within year). Column 1 is the baseline of Table 1. Columns 1–3 estimate cohort effects net of age and calendar-year effects on cohort × age × year cells, weighted by worker-years (the whole sample adds a sex dummy). Columns 4–5 instead subtract from each wage the mean log wage of workers aged 40–45 in the same year, with their composition held at its 2005 value (Appendix B.1); column 5 then removes age effects. At a single age (column 4), cohort and calendar year move together, so year effects cannot be included. Each column then fits a line on each side of the cutoff on the cohort points; the coefficient is the jump at the 2007 cohort. Cohorts 2000–2012, 2006 excluded; the 2012 cohort is observed only up to age 29, so it is not in column 4. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C.6 AKM decomposition of wages

A wage gain can arise because workers from reformed cohorts are paid more in the same firms or because they work in firms that pay all their workers more. To separate the two I estimate a two-way fixed-effects wage equation (Abowd et al., 1999),

$$\log w_{it} = \theta_i + \psi_{J(i,t)} + \mu_t + h(a_{it}) + \epsilon_{it}, \tag{10}$$

where θ_i is a worker effect, $\psi_{J(i,t)}$ the pay premium of the firm $J(i,t)$ that employs worker i in year t, μ_t a year effect and $h(\cdot)$ a quadratic in age. Worker and firm effects are separately identified only among firms linked by workers who move between them, so I estimate equation (10) on the largest connected set of firms and workers.

I then run the discontinuity regression on the wage and on its firm component. Let $\widetilde{w}_{it}$ be the log hourly wage after removing a quadratic in age and year effects. Because $\widetilde{w}_{it} = \hat{\psi}_{J(i,t)} + (\widetilde{w}_{it} - \hat{\psi}_{J(i,t)})$, and because the discontinuity regression is linear and is run on the same worker-years for each term, the wage jump splits exactly into two parts:

$$\Delta^{w} = \Delta^{\psi} + \Delta^{w-\psi}, \qquad \text{firm share} = \frac{\Delta^{\psi}}{\Delta^{w}}. \tag{11}$$

The firm component Δ^{ψ} measures how much of the wage jump comes from working in firms with higher pay premia. The remainder $\Delta^{w-\psi}$ contains the worker effects and everything else. Firm premia are held fixed over the sample period, so the firm share captures moves towards firms that pay more to all their workers, not changes in how firms pay.

C.7 Firm sector decomposition of wages

The AKM decomposition separates the wage effect into worker and firm components, but it does not say in which firms the gain is realised. I therefore split the effect on the log wage by the type of firm the worker is employed in. The split is exact, and each of its pieces is an RD on an outcome defined for every worker, so no piece conditions on where the worker ends up.

Firm types. I classify each employer by its three-digit CAE Rev. 3 sector. QP codes some firms at division level in some years and at group level in others (banks move between 641 and 640, insurers between 651 and 650), so for those divisions I use the division code in every year. I group sectors into four types. *IT and engineering*: computer programming and IT consulting (division 62), information services such as data processing and hosting (63), and architecture and engineering firms (71), the same definition as in Figure 7. *Business and support services*: head offices (701), management consultancy (702), accounting, audit and tax (692), other professional and technical services (749), and divisions 77–82 (rental, staffing, travel, security,

cleaning, office administration and business support). *Banking, insurance and finance*: divisions 64–66. *All other firms.*

Outcomes. Let w_i be the calendar-deflated log hourly wage of worker i, $k(i)$ the type of the firm they work in, and c the mean of w among pre-reform cohorts in the estimation window. For each type k I define

$$y_{ik} = (w_i - c)\,\mathbf{1}\{k(i) = k\}, \tag{12}$$

which is zero for everyone not employed in a type-k firm. Summing over types gives $\sum_k y_{ik} = w_i - c$. The RD estimator is linear in the cohort cell means and c is a constant, so the RD estimates on the y_k add up exactly to the RD on w. The mean of y_k in a cohort is $s_k(\mu_k - c)$, where s_k is the share of the cohort working in type k and μ_k is their mean log wage. Its jump at the cutoff is

$$\Delta\big[s_k(\mu_k - c)\big] = \underbrace{\Delta s_k\,(\mu_k^- - c)}_{\text{more workers in } k} + \underbrace{s_k^-\,\Delta\mu_k}_{\text{higher pay inside } k} + \underbrace{\Delta s_k\,\Delta\mu_k}_{\text{interaction}}, \tag{13}$$

where a superscript $-$ denotes the limit from the pre-reform side. A firm type therefore adds to the wage effect if post-reform cohorts pay more inside it, if more of them work there and it paid above the pre-reform average, or if fewer of them work there and it paid below that average.

The rows say where the reform's wage effect is realised, not how large it would have been had workers not changed sector. Sector is itself an outcome of the reform, so a comparison of wages *within* a sector would compare post-reform workers with pre-reform workers who would have chosen differently. The outcomes y_k avoid that comparison because they are defined for everyone.

C.8 Wage floor and wage cushion decomposition

Following Cardoso and Portugal (2005), I split each worker's wage into the part set by the job title and the part set by the employer. Most Portuguese private-sector workers are covered by collective agreements that fix a minimum rate for each job title (*categoria profissional*). Firms may pay more than that rate but not less. The wage is therefore the sum of a *floor*, the rate attached to the worker's job title, and a *cushion*, the pay the firm adds above it.

Measuring the floor. I do not observe the agreed rates directly, so I recover them from the distribution of pay. Because many workers are paid exactly the rate of their job title, that rate is where wages within the title bunch. For job title c in year y I define the floor $F(c, y)$ as the most common log hourly wage, rounded to 0.01 log points, among all worker-years in *Quadros de Pessoal* that hold the title in that year. Ties go to the lower wage. I use every worker-year in the data, not only the cohorts in the estimation sample, because the floor belongs to the job title rather than to the worker. I compute it only in cells with at least ten worker-years, and I leave out the residual code for workers with no professional category. At ages 28 to 30, 81 percent of worker-years in the sample of Table 4 have a job title with a measured floor.

The decomposition. Let w_{it} be the log hourly wage of worker i in year t, $c(i,t)$ their job title and $I(t)$ the calendar index of Appendix B.1. Then

$$w_{it} - I(t) = \underbrace{\big[F(c(i,t),t) - I(t)\big]}_{\text{floor}} + \underbrace{\big[w_{it} - F(c(i,t),t)\big]}_{\text{cushion}}. \tag{14}$$

As in the other two splits, the discontinuity regression is linear and is run on the same worker-years for each term, so the wage jump is exactly the sum of the jump in the floor and the jump in the cushion, $\Delta^w = \Delta^F + \Delta^{w-F}$ (Table 6). Because the regression includes calendar-year effects, the index $I(t)$ drops out of every term.

D Placebo cutoffs

The main results compare cohorts on either side of 2007. If the cohort series had breaks of this size from time to time, a jump at 2007 would not be evidence of the reform. I therefore move the cutoff to cohorts where nothing happened. I place it at 2002, 2003 and 2004, using only the pre-reform cohorts 2000–2005, and at 2009, 2010 and 2011, using only the post-reform cohorts 2007–2012. I run this for the six main *Quadros de Pessoal* outcomes, each in the design of the main text. University completion, years of university and the three field groups use all ages with age controls (Figures 4 and 6). The log hourly wage uses ages 28 to 30, net of age and calendar-year effects (Table 1, column 1).

Figure 13 and Table 12 show that the jump is specific to 2007. For women, the largest of the six placebo jumps is at most 44 percent of the actual jump, for every outcome. For the field groups and the wage it

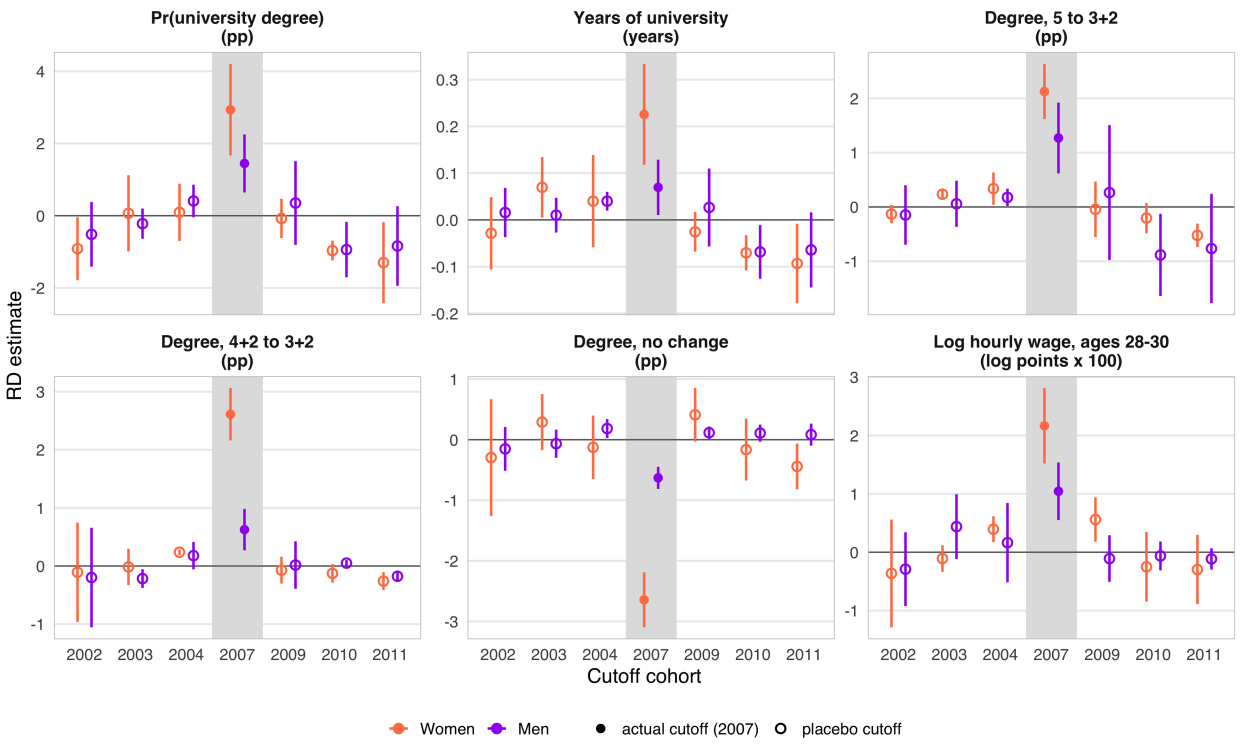


Figure 13: Jumps at the actual and at placebo cutoffs

Notes: Quadros de Pessoal. Each point is the RD estimate at the cutoff cohort on the horizontal axis, with its 95 percent confidence interval. Filled points in the shaded column are the actual cutoff, 2007; hollow points are placebo cutoffs. Placebos at 2002–2004 use only the pre-reform cohorts 2000–2005; placebos at 2009–2011 use only the post-reform cohorts 2007–2012. Every estimate fits a line on each side of its cutoff to the same cohort points. Completion, years of university and the three field groups: all ages, cohort means net of age and age squared, as in Figures 4 and 6. Wage: ages 28–30, cohort effects net of age and calendar-year effects, as in Table 1, column 1. Shares in percentage points; wage in log points × 100.

is at most a quarter. For men, the largest placebo is at most 70 percent of the actual jump for five of the six outcomes, and 42 percent for the wage. The exception is men's years of university. There the actual jump is small, 0.07 years, and the 2010 placebo is as large but of the opposite sign. Five of the 72 placebo estimates are significant at the 5 percent level, close to the number expected by chance. These standard errors are not reliable, because each placebo rests on two to four cohorts on each side. What matters is the size of the placebo jumps relative to the actual one. The placebos among post-reform cohorts lean negative for completion, years of university and the 5 to 3+2 group, because the post-reform series flattens in the latest cohorts. These placebos have the opposite sign to the actual jump, so they cannot account for it.

Table 12: Effect of the reform at the actual and at placebo cutoffs

	(1)	(2)	(3)	(4)	(5)	(6)
			Degree, by what the reform did to the field			
Cutoff cohort	Pr(university)	Years of univ.	5 to 3+2	4+2 to 3+2	No change	Log wage
Panel A. Women						
2007 (actual)	0.029***	0.226***	0.021***	0.026***	−0.026***	0.022***
	(0.006)	(0.055)	(0.003)	(0.002)	(0.002)	(0.003)
2002 (placebo)	−0.009	−0.029	−0.001	−0.001	−0.003	−0.004
	(0.004)	(0.039)	(0.001)	(0.004)	(0.005)	(0.005)
2003 (placebo)	0.001	0.070	0.002**	0.000	0.003	−0.001
	(0.005)	(0.033)	(0.001)	(0.002)	(0.002)	(0.001)
2004 (placebo)	0.001	0.040	0.003	0.002***	−0.001	0.004*
	(0.004)	(0.050)	(0.002)	(0.000)	(0.003)	(0.001)
2009 (placebo)	−0.001	−0.026	0.000	−0.001	0.004	0.006
	(0.003)	(0.022)	(0.003)	(0.001)	(0.002)	(0.002)
2010 (placebo)	−0.010**	−0.070*	−0.002	−0.001	−0.002	−0.002
	(0.001)	(0.019)	(0.001)	(0.001)	(0.003)	(0.003)
2011 (placebo)	−0.013	−0.093	−0.005**	−0.003*	−0.004	−0.003
	(0.006)	(0.043)	(0.001)	(0.001)	(0.002)	(0.003)
Largest placebo / actual	0.44	0.41	0.25	0.10	0.17	0.26
Panel B. Men						
2007 (actual)	0.014***	0.070**	0.013***	0.006***	−0.006***	0.010***
	(0.004)	(0.030)	(0.003)	(0.002)	(0.001)	(0.003)
2002 (placebo)	−0.005	0.016	−0.001	−0.002	−0.002	−0.003
	(0.005)	(0.027)	(0.003)	(0.004)	(0.002)	(0.003)
2003 (placebo)	−0.002	0.010	0.001	−0.002	−0.001	0.004
	(0.002)	(0.019)	(0.002)	(0.001)	(0.001)	(0.003)
2004 (placebo)	0.004	0.040*	0.002	0.002	0.002	0.002
	(0.002)	(0.010)	(0.001)	(0.001)	(0.001)	(0.003)
2009 (placebo)	0.004	0.027	0.003	0.000	0.001	−0.001
	(0.006)	(0.043)	(0.006)	(0.002)	(0.001)	(0.002)
2010 (placebo)	−0.009	−0.069	−0.009	0.000	0.001	−0.001
	(0.004)	(0.029)	(0.004)	(0.000)	(0.001)	(0.001)
2011 (placebo)	−0.008	−0.064	−0.008	−0.002**	0.001	−0.001
	(0.006)	(0.041)	(0.005)	(0.000)	(0.001)	(0.001)
Largest placebo / actual	0.65	0.98	0.70	0.35	0.29	0.42

Notes: Quadros de Pessoal. RD estimates at the cutoff cohort in each row, from a line on each side of the cutoff fitted to the cohort points. The 2007 row is the main estimate (cohorts 2000–2012, 2006 excluded). Placebo rows at 2002–2004 use only the pre-reform cohorts 2000–2005; rows at 2009–2011 use only the post-reform cohorts 2007–2012; no cohort is excluded. Columns 1–5: all ages, cohort means net of age and age squared. Column 1: the highest completed level is a university degree. Column 2: years of university, zero for non-graduates. Columns 3–5: a university degree in each group of fields, as in Figure 6. Column 6: log real hourly wage at ages 28–30, cohort effects net of age and calendar-year effects, as in Table 1, column 1. "Largest placebo / actual": the largest of the six placebo estimates in absolute value, divided by the absolute value of the actual jump. Placebo estimates rest on two to four cohorts on each side, so their standard errors are imprecise. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

E Degrees by field of study

Figure 14 splits the field groups of Section 5.2 into single fields, with STEM split into engineering and computing and into the sciences and mathematics. The top row is the effect on holding any university degree.

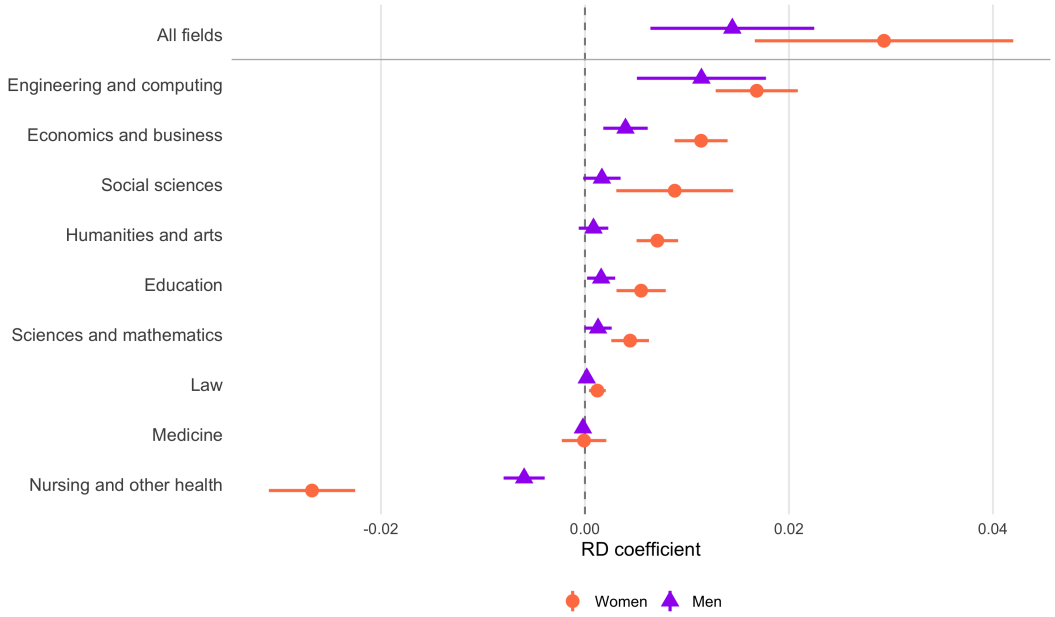


Figure 14: Effect of the reform on university degrees, by field of study

Notes: *Quadros de Pessoal*. Each point is a separate RD estimate of the share of the cohort holding a university degree in the field, with a 95 percent confidence interval. All fields: any university degree, as in Figure 4. Outcomes net of a quadratic in age; cohort-by-sex means, linear on each side of the cutoff, cohorts 2000–2012 (2006 excluded). Fields are ISCED 1997 groups; a missing field is filled in from the occupation. Engineering and computing: computing, engineering, manufacturing and processing, and architecture and construction (ISCED 48, 52, 54, 58). Sciences and mathematics: life and physical sciences, mathematics and statistics (ISCED 42, 44, 46). The two add up to STEM; graduates whose field was filled in from the occupation (8 percent of STEM) count as sciences if they ever work as scientists and as engineering otherwise. Economics and business: business and administration (ISCED 34). Medicine: graduates of the health group who work as medical doctors. Nursing and other health: the rest of the health group (health, ISCED 72, together with agriculture and veterinary science, ISCED 62 and 64), mostly nursing.

F Results in other data sources

This appendix reports the main education results in the three Portuguese data sources: *Quadros de Pessoal*, the Labour Force Survey and the Education and Training Survey (IEFA). Table 13 covers university completion and years of university (Section 5.1); Table 14 covers degrees by what the reform did to the field (Section 5.2).

Table 13: Effect of the reform on university completion and years of university, three data sources

	Highest education is a university degree			Years of university		
	QP	LFS	IEFA	QP	LFS	IEFA
Women	0.029***	0.037***	0.056	0.226***	0.185***	0.281
	(0.006)	(0.004)	(0.041)	(0.055)	(0.024)	(0.263)
Baseline	0.340	0.229	0.287	2.758	1.938	1.811
Men	0.014***	0.012***	0.049	0.070**	0.070***	0.147
	(0.004)	(0.003)	(0.053)	(0.030)	(0.020)	(0.357)
Baseline	0.204	0.133	0.121	1.845	1.608	1.730
p-value, women = men	0.052	0.000	0.921	0.013	0.000	0.762

Notes: RD estimates at the 2007 cohort, excluding the 2006 cohort, with separate linear trends in the running variable on each side and age controls. QP: the estimates of Figure 4, from cohort means (Section 4); baseline is the mean over the pre-reform cohorts. LFS: individual-level regression with conventional standard errors; IEFA: weighted cohort means; for both, baseline is the mean in the last pre-reform cohort. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Observations (women / men): QP 6,011,539 / 6,636,952 worker-years; LFS 187,114 / 191,608 person-quarters; IEFA 3,648 / 3,532 persons (weighted). Years of university are zero for non-graduates; in QP they are proxied by the age at which a worker is first observed with their highest degree, minus 18 (Appendix B.1).

Table 14: Effect of the reform on university degrees by what the reform did to the field, three data sources

	5 to 3+2			4+2 to 3+2			No change		
	QP	LFS	IEFA	QP	LFS	IEFA	QP	LFS	IEFA
Women	0.021***	0.008***	0.024	0.026***	0.037***	0.024	−0.026***	−0.010***	−0.001
	(0.003)	(0.002)	(0.021)	(0.002)	(0.004)	(0.037)	(0.002)	(0.003)	(0.025)
Baseline	0.064	0.027	0.022	0.122	0.111	0.189	0.092	0.062	0.076
Men	0.013***	0.002	0.017	0.006***	0.009***	0.032	−0.006***	−0.003**	0.002
	(0.003)	(0.002)	(0.014)	(0.002)	(0.002)	(0.035)	(0.001)	(0.001)	(0.017)
Baseline	0.102	0.049	0.054	0.059	0.051	0.063	0.025	0.013	0.004
p-value, women = men	0.042	0.077	0.802	0.000	0.000	0.872	0.000	0.012	0.908

Notes: RD estimates at the 2007 cohort, excluding the 2006 cohort, with separate linear trends in the running variable on each side and age controls. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The outcome is the share of the cohort whose highest completed level is a university degree in each group of fields. QP: the estimates in Figure 6, from cohort means; field is taken from occupation when missing; the social sciences are in none of the three groups; baseline is the mean over pre-reform cohorts. LFS and IEFA: social sciences, business and law share one code and are in the 4+2 to 3+2 group, so "no change" is health only; baseline is the mean in the last pre-reform cohort. Observations (women / men): QP 6,011,539 / 6,636,952 worker-years; LFS 161,506 / 167,591 person-quarters; IEFA 3,648 / 3,532 persons (weighted).

G Field choices in ministry records

The field results in Section 5.2 come from people observed at work. A shift in degrees among workers could in principle reflect who ends up in private-sector employment rather than what students chose to study. This appendix checks the shift in records kept by the Ministry of Higher Education, which observe students when they apply, when they enter and when they graduate. These records cover public and private institutions and every student, whatever they later do. They are counts by calendar year, so they do not allow the regression discontinuity design of Section 4. I compare the years before and after the reform and read the results as descriptive evidence. Because the age at entry did not change with the reform (Figure 3b), the year of entry corresponds closely to the cohort in the main analysis.

Data. I use three records. The first is the national contest for public institutions, run by the Directorate-General for Higher Education (DGES), for 2004 to 2009. For each programme it reports the applications received and the students placed in the first round, both by sex. An applicant can list up to six programmes, and every listed option counts as an application, so applications measure demand while placements are limited by the number of places. The second record counts all students enrolled in the first year of a first degree for the first time, in public and private institutions, for 2004/05 to 2010/11. The third counts the first degrees awarded each year from 2006/07 to 2014/15. Both come from the ministry's statistics office (GPEARI, later DGEEC) and report each programme by sex. I group programmes as in Figure 7: engineering covers engineering, manufacturing and processing, and architecture and construction; economics and business covers economics and business and administration; nursing covers the programmes named *Enfermagem*. For each source and sex I compute the share of all applications, placements or first-year students that falls in each group. Before the reform means 2004 and 2005, and after means 2007 to 2009. I leave out 2006, as in the main analysis, because 43 percent of programmes had already converted to the new structure that year (Section 2).

Table 15 shows the same shift as *Quadros de Pessoal* in every record. After the reform, a larger share of women applied to and enrolled in engineering. A larger share of applications from both women and men went to economics and business. A much smaller share of women and men applied to and enrolled in nursing.

Table 15: Share of applications, placements and first-year students by field, before and after the reform

		Women			Men		
		Before	After	Change	Before	After	Change
Engineering	Applications (DGES)	8.7	10.9	+2.2	24.6	24.9	+0.4
	First-year students	7.6	9.6	+2.0	27.6	32.2	+4.6
Economics and business	Applications (DGES)	12.5	15.0	+2.5	14.2	17.2	+3.0
	First-year students	15.7	18.7	+3.0	18.5	19.7	+1.2
Nursing	Applications (DGES)	10.9	6.6	−4.3	3.8	1.9	−1.8
	First-year students	8.8	6.5	−2.3	2.7	1.7	−1.0

Notes: Percent of all applications or first-year students of the same sex. Before: average of 2004 and 2005. After: average of 2007, 2008 and 2009. Change in percentage points, computed before rounding.

H Graduate Oversupply

The reform raised the number of new graduates in engineering and in economics and business. If the labour market could not absorb them, post-reform graduates would start in worse jobs or in jobs outside their field. This appendix checks two such margins. Graduates more often started as trainees, but only for a year. They were not less likely to work in their field.

Sample. I use graduate entrants: workers in their first year in QP with a university degree, hired in the last twelve months, aged 21–28. Cohorts are the worker's modal cohort, and the 2006 cohort is excluded. I follow each entrant's main job in the year of the first job and in each of the next three years. In year k the sample keeps entrants whose first job came by age $29 - k$, so every cohort is observed k years later. All estimates are cohort RDs, as in the rest of the paper.

H.1 More trainees, but only for a year

QP records each worker's level in the firm, and the lowest level is reserved for trainees and apprentices. Before the reform, 12.0 percent of female and 11.5 percent of male graduate entrants started on this grade. Table 16 shows that this share jumps at the cutoff, by 3.1 percentage points for women and 2.5 for men. Most of the extra traineeships carry no recorded pay and fall in 2012–16. This is consistent with subsidised professional internships.

The jump, however, does not last. One year after the first job, the jump is 0.4 percentage points for women and not significant. For men it is −0.9 points. Two and three years after the first job, the jump is zero or negative for both.

H.2 No change in field match where graduates rose

The match rate is the share of a field's graduate entrants whose main job is in the sector, or the occupation, that matches the field. I use the three fields and matches of Figure 7. Engineering graduates match IT and engineering firms, and engineering and ICT professionals and technicians. Economics and business graduates match finance and business services, and business and finance professionals. Health graduates match health

Table 16: Trainee grade by years since the first job, all graduate entrants (percentage points)

	First job	1 year later	2 years later	3 years later
Women				
RD jump	3.1***	0.4	-0.2	-0.5
	(0.9)	(0.3)	(0.3)	(0.3)
Pre-reform baseline	12.0	5.2	3.8	2.7
Men				
RD jump	2.5**	-0.9***	-1.1***	-0.9**
	(1.0)	(0.3)	(0.3)	(0.3)
Pre-reform baseline	11.5	4.3	3.0	2.1
Graduate entrants	316,640	316,640	284,324	245,563

Notes: *Quadros de Pessoal*, graduate entrants of all fields (see text). Outcome: main job on the trainee grade (level 8) in the year of the first job and one to three years later; zero for an entrant in QP on another grade. Cohort RD estimates with standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

firms, and health professionals and associate professionals. The rate is measured among entrants who are in QP that year.

Table 17 reports the jump in the match rate at the first job and in the next three years. In engineering, no estimate is significant, for either sex or either match. All lie within about 3 percentage points of zero. In economics and business, the occupation match does not fall. The sector match is 3 to 6 percentage points lower, on a baseline of 34 to 37 percent, but only two of the eight estimates are significant, both at the 10 percent level.

The match rate falls only in health, the field whose number of graduates fell. Health graduates more often work outside health firms, by 8 to 9 percentage points at the first job and by 13 to 16 points three years later. They also more often work outside health occupations, by 10 to 14 points three years later.**[TK: why health graduates left health jobs: public hospital hiring freeze 2011–14 and nurse emigration? see the health group check]** An excess of graduates cannot explain a fall in a field whose graduates became fewer. In the two fields that gained graduates, the labour market absorbed them in their own field.

Table 17: Field match rate by years since the first job, by field (percentage points)

Field	Match by	First job	1 year later	2 years later	3 years later	Baseline
Women						
Engineering	Sector	-0.7	2.0	0.5	0.9	24.2
		(1.2)	(1.8)	(0.9)	(2.0)	
	Occupation	-0.3	-0.4	1.3	2.6	43.3
		(1.7)	(2.4)	(1.5)	(3.0)	
Economics and business	Sector	-2.6	-2.7	-3.9*	-2.5	33.7
		(1.4)	(1.8)	(1.9)	(1.7)	
	Occupation	1.3	0.5	-0.7	-2.2	22.5
		(1.5)	(1.4)	(1.1)	(1.5)	
Health	Sector	-9.3**	-11.1**	-11.2**	-12.6**	49.0
		(4.0)	(4.3)	(3.7)	(4.1)	
	Occupation	-3.6**	-7.5***	-8.5***	-9.8***	79.4
		(1.2)	(1.2)	(1.3)	(1.6)	
Men						
Engineering	Sector	-2.3	-2.7	-3.1	-2.6	29.3
		(2.2)	(2.4)	(2.5)	(2.7)	
	Occupation	0.4	-0.4	-1.6	-1.7	65.3
		(0.9)	(1.2)	(1.9)	(1.2)	
Economics and business	Sector	-3.6	-6.2*	-5.6	-3.2	37.0
		(2.2)	(3.1)	(3.5)	(3.3)	
	Occupation	2.4	2.7*	0.6	1.9	23.4
		(1.8)	(1.4)	(2.1)	(3.2)	
Health	Sector	-7.6**	-10.3**	-12.5**	-16.1***	48.9
		(3.0)	(3.4)	(4.1)	(3.9)	
	Occupation	-2.4	-9.0**	-8.0**	-13.8***	78.3
		(2.4)	(3.7)	(3.4)	(3.8)	

Notes: Quadros de Pessoal, graduate entrants of each field who are in QP that year (see text). Outcome: main job in the sector or the occupation that matches the field. Fields: engineering (ISCED 52, 54, 58), economics and business (34), health (72). Sectors (CAE Rev. 3): 62, 63, 71; 64–66, 69, 70; 86. Occupations (ISCO-08): 214, 215, 251, 252, 311; 241–243; 221–226, 321–325. Cohort RD estimates with standard errors in parentheses. Baseline: pre-reform match rate at the first job (percent). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

I The model: details

I.1 Solution

Timing. A student who enters a field studies the first year. At the end of each year she learns whether she passed it, receives the signal if it is the end of her third year, updates her belief, and then chooses whether to leave or to continue. She graduates when her credits reach the full degree. Enrolment is capped at three years beyond the full degree.

Beliefs. A good match passes every year; a poor match fails each year with probability q. After t years with c passed and no signal yet, a student who started with belief p holds

$$b_t(c) = \frac{p\,\mathbf{1}[c=t]}{p\,\mathbf{1}[c=t] + (1-p)\,q^{\,t-c}(1-q)^c}, \tag{15}$$

so a single failed year reveals a poor match and each year passed raises the belief. The signal at the end of the third year is good with probability $\frac{1}{2}(1+\rho)$ for a good match and $\frac{1}{2}(1-\rho)$ for a poor match, with $\rho = 0.99$, and the belief is updated by Bayes' rule. The model is solved exactly on the tree of pass–fail records and signals, on a grid of starting beliefs.

Values. I solve (5)–(6) backwards from the enrolment cap, at which every student still enrolled must leave, over years, credits and signal paths, for every starting belief on the grid.

Entry. Entry follows (7), with beliefs from (3). I simulate 20,000 students from the PISA 2003 sample with its weights and draw ε_{if} once; the argmax is smoothed by a softmax with a temperature of 0.0005 yearly wages, far below the cost of one year of study. The field shifts a_f solve a fixed point: at the pre-reform menus, the model's share of the cohort holding a degree in each field equals the data's. Because beliefs are correct, a higher a_f means more good matches in the field, not a taste for it.

I.2 Fixed parameters

Table 18 lists what I set outside the model. Wages are in units of the wage of workers without university, from the occupations of credential holders in Quadros de Pessoal. The pre-reform shares are those of Figure 6, averaged over women and men; together with the share without a university degree they cover the whole cohort.

Table 18: Fixed parameters

	Intermediate credential	Shorter first degree	No change
Wages, good match / poor match			
Intermediate credential (three years)	1.59 / 1.59	–	–
Full degree	1.83 / 1.59	1.67 / 1.67	1.65 / 1.65
Years to the credentials			
Before the reform	five	four	four
After the reform	three and five	three	four
Pre-reform share of the cohort with a degree (%)	8.3	9.1	5.9
Field shift a_f (at the estimates)	−5.49	−5.36	−6.35

Notes: Common to all fields: discount factor $\beta = 0.95$; working life ends 38 years after entry; yearly direct cost $k = 0.1$; one signal of precision $\rho = 0.99$ at the end of the third year; dislike of risk $\gamma = 6$; enrolment capped at three years beyond the full degree. Starting beliefs from the PISA 2003 predicted probability of expecting to complete university (mean 0.51), with the field shifts and the spread s. Wages: engineers (1.83), technical engineers (1.59), non-STEM graduates (1.67) and nurses (1.65). The model has no master's after the shorter first degree.

I.3 Estimation and identification

Estimator. I minimise $\sum_m \left[(\hat{m}_m - m_m(\theta))/se_m\right]^2$ over $\theta = (s, q)$ with Nelder–Mead from two starting points, with bounds imposed by a logistic transformation. The five-year moment enters only when the model falls short of it. The loss at the estimate is 0.38 for five moments and two parameters; neither parameter ends at a bound. Standard errors in Table 19 are $(G'WG)^{-1}$, with G the Jacobian of the targeted moments at the estimate (central differences, field shifts re-set at every evaluation) and W the inverse of the moments' variances.

Table 19: Estimated parameters

Parameter	Estimate	(SE)
Spread of beliefs s	4.33	(1.38)
Poor match's chance of failing a year q	0.537	(0.038)

Notes: Local and approximate standard errors; the policy functions have kinks in the parameters.

Identification. The chance of failing a year q is pinned mainly by the pre-reform dropout level: a higher q means more poor matches fall behind and leave. It also moves the field responses, since slower progress makes the saved year and the intermediate credential worth more to students who fall behind. The spread s is pinned by the size of the field responses. A wider spread leaves fewer students close to indifferent between a field and working, so entry responds less to the reform.

Dislike of risk. I fix γ because the five moments cannot pin it together with the other parameters. Table 20 refits s and q at other values. The fit is best at $\gamma = 6$ and close at $\gamma = 4$ and $\gamma = 8$. With little or no dislike of risk, the model overstates the pre-reform dropout level, and good matches completing five years barely rise. Without the dislike of risk, the intermediate credential is worth most to the least confident students: at the estimated s and q it is worth 8.0 percent of a yearly wage at a belief of 0.3, and its value falls steadily with the belief. The dislike of risk shifts its value towards students who expect to finish but fear they might not.

Table 20: The fit at other values of γ

γ	0	2	4	6	8
Loss	50.4	16.6	0.83	0.38	0.81
s	6.00	5.62	5.08	4.33	3.32
q	0.64	0.60	0.56	0.54	0.54
Δ degrees, intermediate credential (pp)	1.38	1.60	1.77	1.62	1.68
Δ degrees, shorter first degree (pp)	2.10	1.92	1.64	1.67	1.64
Δ good matches completing five years (pp)	−0.03	0.24	0.54	0.97	1.15
Δ all five-year graduates (pp)	−0.11	0.15	0.48	0.94	1.15
Entrants leaving with no degree, before (%)	26.9	22.8	19.4	16.4	14.3
Value of the intermediate credential (% of a yearly wage)	1.5	1.9	2.2	2.7	2.8
Value of the saved year (% of a yearly wage)	7.5	7.2	6.7	6.1	5.1
Wage effect (log points)	1.8	1.8	1.8	1.8	1.8

Notes: At each γ, s and q are re-estimated on the same five moments and the field shifts re-set. Values: average gain to students who would have entered the field before the reform, every year for life.

I.4 All counterfactuals

Table 21 adds outcomes and two rows to Table 8. When students learn their match after one year, the intermediate credential draws in no one and dropout barely changes. The last two rows switch off the dislike of risk at the estimated spread of beliefs, holding the field shifts fixed or re-setting them to the pre-reform shares. In both, entry rises by more than half of the cohort. At the estimated spread, the dislike of risk is what keeps entry from responding strongly, so this world says little about the channels, and the text uses the timing of learning instead.

Who stops at three years. Following the same student along the same record of passes, fails and signals under both menus, 0.06 percent of the cohort complete five years before the reform and stop at three after it; no student moves the other way. Half of them are poor matches who passed their first three years. Before the reform they stayed for the five-year degree, which paid them no more than three years would have; after it they save two years. The other half are good matches who received a misleading signal at the end of the third year. The flow is small because a poor match fails about one year in two, so almost every student who reaches the end of the third year without failing is a good match. In the same comparison, 0.30 percent of the cohort leave with nothing before the reform and with the intermediate credential after it.

Table 21: The reform's effect with one feature changed: all outcomes

	Intermediate credential	Shorter first degree	No change	University entry	Dropout	Five-year, good	All five-year	New entrants	Wage effect
The reform	1.62	1.67	−0.28	4.60	1.59	0.97	0.94	2.68	1.8
Intermediate credential only	1.95	−0.11	−0.09	2.59	0.84	1.28	1.25	2.86	1.1
Shorter first degree only	−0.28	1.79	−0.18	2.14	0.80	−0.28	−0.28	0.00	0.7
Match learned after year 1	−0.25	0.55	−0.13	0.49	0.31	−0.27	−0.27	0.00	0.1
No dislike of risk, shifts fixed	10.75	26.66	−1.22	57.40	21.22	0.43	0.09	13.47	17.7
No dislike of risk, shifts re-set	10.83	30.06	−0.93	65.80	25.84	0.63	0.32	13.46	20.2

Notes: Changes in percentage points of the cohort; wage effect in log points. Dropout: share of the cohort leaving with no degree. Five-year, good: good matches completing the five-year degree. All five-year: everyone completing it. New entrants: students who enter the field with the intermediate credential only after the reform. Each row holds the estimated parameters fixed, changes one feature, and compares the pre- and post-reform menus within that world.

I.5 Decomposing the wage effect

Table 22 splits the model's wage effect by the route students end up on. More graduates with a shorter first degree add 1.0 point. In the field with the intermediate credential, more five-year graduates add 0.7 points and students leaving with the new intermediate credential 0.35. Fewer graduates in the unchanged fields subtract 0.15. Each route pays the same wage before and after the reform, so the whole effect comes from where students end up.

Table 22: Where the model's wage effect comes from

Route	Share of cohort (%) before	Share of cohort (%) after	Mean wage before	Mean wage after	Contribution (% of mean wage)
Never enter	72.2	67.6	1.00	1.00	0
Intermediate-credential field, leave with nothing	1.6	2.4	1.00	1.00	0
intermediate credential	0.0	0.7	–	1.59	0.35
five-year degree	8.3	9.3	1.83	1.83	0.67
Shorter-first-degree field, leave with nothing	1.8	2.6	1.00	1.00	0
degree	9.1	10.8	1.67	1.67	0.96
Unchanged field, leave with nothing	1.2	1.2	1.00	1.00	0
degree	5.9	5.6	1.65	1.65	−0.15
Total					1.82

Notes: Contribution of route j: $[\, s_j^{\text{post}}(w_j^{\text{post}} - 1) - s_j^{\text{pre}}(w_j^{\text{pre}} - 1)\,]/\bar{W}^{\text{pre}}$, with $\bar{W}^{\text{pre}} = 1.168$; the total is the percent change in the cohort's mean wage (1.80 log points).

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